

Generic properties of Steklov eigenfunctions on graphs

Kun Yang*

Abstract

Weak unique continuation fails for the discrete Steklov problem. For generic positive edge weights, every nonconstant eigenfunction is injective on the boundary, and its harmonic extension is injective on the B -core. The extension is then a vertex Morse function on the core in the sense of Knill, and the index sum equals $\chi(G^B)$. If the support graph of S is a path, a generic eigenfunction belonging to λ_j has $j + 1$ local extrema; if it is a cycle, the count is $2\lceil j/2 \rceil$; if it is a tree, the local extrema of λ_1 are the leaves. If every interior component meets B in at most two vertices, a generic eigenfunction belonging to a positive eigenvalue has as many nodal domains on V as on G_S .

1 Introduction

Generic properties of eigenfunctions go back to Uhlenbeck [2]. On a compact Riemannian manifold with boundary, Wang [1] proved that, for a generic conformal class, Steklov eigenfunctions are Morse on the boundary: the differential is transverse to the zero section of $T^*(\partial M)$. The argument uses weak unique continuation and a density lemma for the first variation of the Dirichlet-to-Neumann map.

The discrete Steklov problem on a finite graph, with a designated boundary $B \subset V$, was studied by Hua–Huang–Wang [3]. Chen–Hua–Shi [4] established that, for generic positive edge weights, every Steklov eigenvalue is simple and every eigenfunction is nowhere zero on B . Simplicity and nonvanishing do not yield injectivity: the constant eigenfunction belonging to $\lambda_0 = 0$ takes the same value at every boundary vertex whenever $|B| \geq 2$. Weak unique continuation likewise fails, since a singleton is open in the discrete topology. Wang’s density lemma therefore does not transfer. The proofs below use resultants of characteristic polynomials and inductive constructions on trees.

For generic weights, every nonconstant eigenfunction is injective on B . The same compression excludes $\alpha^\top f = 0$ whenever $\alpha \in C^B$ is not parallel to $\mathbf{1}$, for any finite $C \subset \mathbb{R}$; in particular $2f(i) \neq f(j)$, $f(i) + f(j) \neq 0$, and $2f(j) \neq f(i) + f(k)$. A vertex or edge lies in the B -core G^B when it belongs to a simple path with both endpoints in B . Interior leaves and hanging blocks lie outside G^B , and the extension is constant on each such block for every $w > 0$. On G^B the extension is injective and nowhere zero, hence a vertex Morse function in the sense of Knill [8], and

$$\sum_{v \in V(G^B)} i_u(v) = \chi(G^B).$$

If G admits a spanning tree whose leaves all lie in B , then $G^B = G$. The converse fails on C_4 with B a pair of opposite vertices.

Closed stars in $G[B]$ and in G_S lie in B , so injectivity on B yields vertex Morse on both. If G_S is a path, the spectrum is simple for every $w > 0$, and a generic eigenfunction for λ_j has $j + 1$ local

*School of Mathematical Sciences, Nankai University, Tianjin 300071, China. 2313878@mail.nankai.edu.cn.

extrema. If G_S is a cycle, the count is $2\lceil j/2 \rceil$. If G_S is a tree, the local extrema of λ_1 are exactly the leaves. If every interior component meets B in at most two vertices, then for every positive eigenvalue the number of nodal domains of the harmonic extension on V equals the number of nodal domains of f on G_S ; in particular λ_1 has two.

The discrete Steklov problem is recalled next, and vertex Morse functions in Subsection 1.2. Section 2 states the main results. Section 3 collects first variation formulae and the Chen–Hua–Shi input. The remaining statements are proved in Section 4.

1.1 The discrete Steklov problem

Let $G = (V, E)$ be a finite simple undirected graph, and let $B \subset V$ be a nonempty subset, referred to as the *boundary*. The complement $B^c = V \setminus B$ is the *interior*. A *positive edge weight* is a function $w : E \rightarrow \mathbb{R}_+$. Functions on G are identified with vectors in $\mathbb{R}^V$. If G is disconnected, the Steklov problem decomposes over those connected components that meet B ; components disjoint from B do not contribute. Henceforth G is assumed connected.

Definition 1.1. For $u \in \mathbb{R}^V$, the *weighted graph Laplacian* is

$$(\Delta u)(x) = \sum_{\{x,y\} \in E} w(x,y)(u(x) - u(y)), \quad x \in V.$$

The associated *Laplace matrix* L has entries

$$L_{xy} = \begin{cases} \sum_{z \sim x} w(x,z), & y = x, \\ -w(x,y), & \{x,y\} \in E, \\ 0, & \text{otherwise.} \end{cases}$$

The matrix L is symmetric and diagonally dominant. On the boundary the same weighted differences are written as an *outward normal derivative*,

$$(\partial_\nu u)(x) = \sum_{\{x,y\} \in E} w(x,y)(u(x) - u(y)), \quad x \in B.$$

Partitioning L according to $V = B^c \sqcup B$ yields

$$L = \begin{pmatrix} L_{[B^c]} & L_{[B^c, B]} \\ L_{[B, B^c]} & L_{[B]} \end{pmatrix}.$$

Definition 1.2. A pair $(\lambda, u) \in \mathbb{R} \times \mathbb{R}^V \setminus \{0\}$ is a *Steklov eigenpair* if

$$\begin{cases} (\Delta u)(x) = 0, & x \in B^c, \\ (\partial_\nu u)(x) = \lambda u(x), & x \in B. \end{cases} \quad (1.1)$$

Lemma 1.3. If G is connected and $B \neq \emptyset$, then $L_{[B^c]}$ is positive definite, hence invertible. If $B^c = \emptyset$, the block is the 0×0 matrix, with determinant 1.

Proof. If $B^c = \emptyset$, the determinant is 1 by convention. Now assume $B^c \neq \emptyset$. For $x \in \mathbb{R}^{B^c}$, extend x by zero on B to obtain $\tilde{x} \in \mathbb{R}^V$. Then

$$x^\top L_{[B^c]} x = \tilde{x}^\top L \tilde{x} = \sum_{\{a,b\} \in E} w(a,b)(\tilde{x}(a) - \tilde{x}(b))^2.$$

If the left-hand side vanishes, then $\tilde{x}$ is constant along every edge. Connectedness of G forces $\tilde{x}$ to be constant on V . The condition $\tilde{x}|_B = 0$ then yields $\tilde{x} \equiv 0$. Thus the quadratic form is positive definite. $\square$

Definition 1.4. For $f \in \mathbb{R}^B$, the *harmonic extension* $h_f \in \mathbb{R}^V$ is the unique solution of the discrete Dirichlet problem

$$\begin{cases} (\Delta h_f)(x) = 0, & x \in B^c, \\ h_f(x) = f(x), & x \in B. \end{cases} \quad (1.2)$$

Equivalently, $L_{[B^c]}(h_f)_{B^c} + L_{[B^c, B]}f = 0$. By Lemma 1.3,

$$(h_f)_{B^c} = -L_{[B^c]}^{-1} L_{[B^c, B]}f. \quad (1.3)$$

The map $f \mapsto h_f$ is linear, and $h_{\mathbf{1}_B} = \mathbf{1}_V$.

Definition 1.5. The *Dirichlet-to-Neumann operator* is the linear map

$$\Lambda_w : \mathbb{R}^B \rightarrow \mathbb{R}^B, \quad f \mapsto \partial_\nu h_f, \quad (1.4)$$

where h_f is the harmonic extension of f . Pointwise,

$$(\Lambda_w f)(x) = \sum_{\{x, y\} \in E} w(x, y) (h_f(x) - h_f(y)), \quad x \in B. \quad (1.5)$$

Splitting the neighbours of x according to whether they lie in B or in B^c gives

$$(\Lambda_w f)(x) = \sum_{\substack{y \in B \\ \{x, y\} \in E}} w(x, y) (f(x) - f(y)) + \sum_{\substack{z \in B^c \\ \{x, z\} \in E}} w(x, z) (f(x) - h_f(z)). \quad (1.6)$$

In matrix form, $\partial_\nu h_f = L_{[B]}f + L_{[B, B^c]}(h_f)_{B^c}$. Substituting (1.3) yields

$$\Lambda_w f = S f, \quad S = L_{[B]} - L_{[B, B^c]} L_{[B^c]}^{-1} L_{[B^c, B]}. \quad (1.7)$$

The matrix S is the *Steklov matrix*, the Schur complement of $L_{[B^c]}$ in L . By Cramer's rule,

$$S = \frac{L_{[B]} \det L_{[B^c]} - L_{[B, B^c]} \operatorname{adj}(L_{[B^c]}) L_{[B^c, B]}}{\det L_{[B^c]}},$$

so each entry of S is a rational function of the edge weights, with denominator $\det L_{[B^c]}$. The latter is strictly positive for $w > 0$ by Lemma 1.3.

The Steklov problem (1.1) is equivalent to the eigenvalue problem

$$\Lambda_w f = \lambda f, \quad f = u|_B \neq 0, \quad (1.8)$$

with $u = h_f$. If $B^c = \emptyset$, then $h_f = f$ and Λ_w coincides with the graph Laplacian.

For distinct boundary vertices $x, y \in B$,

$$-S_{xy} = w(x, y) + \sum_{z, t \in B^c} w(x, z) (L_{[B^c]}^{-1})_{zt} w(t, y). \quad (1.9)$$

For an interior vertex $x \in B^c$, (1.3) gives

$$h_f(x) = e_x^\top (h_f)_{B^c} = -e_x^\top L_{[B^c]}^{-1} L_{[B^c, B]}f.$$

Since L is symmetric, $L_{[B, B^c]} = L_{[B^c, B]}^\top$, and therefore

$$h_f(x) = \left(-L_{[B, B^c]} L_{[B^c]}^{-1} e_x \right)^\top f = p_x^\top f,$$

where

$$p_x = -L_{[B, B^c]} L_{[B^c]}^{-1} e_x \in \mathbb{R}^B.$$

Thus, for every $x \in B^c$,

$$h_f(x) = p_x^\top f. \quad (1.10)$$

The y -component of p_x is the value at x of the harmonic extension of δ_y . Since the harmonic extension of $\mathbf{1}$ is $\mathbf{1}$, one has $\mathbf{1}_B^\top p_x = 1$.

Lemma 1.6 (Proposition 2.1, [3]). *For every $f \in \mathbb{R}^B$,*

$$f^\top S f = h_f^\top L h_f = \sum_{\{x, y\} \in E} w(x, y) (h_f(x) - h_f(y))^2 \geq 0. \quad (1.11)$$

Proposition 1.7 (Propositions 2.1 and 3.2, [3]). *The matrix S is real symmetric and positive semidefinite. If G is connected, then $\ker S = \text{span}\{\mathbf{1}\}$ and*

$$\text{spec}(S) = \{0 = \lambda_0 < \lambda_1 \leq \dots \leq \lambda_{|B|-1}\}.$$

Every eigenfunction belonging to a nonzero eigenvalue satisfies $\sum_{x \in B} f(x) = 0$.

Lemma 1.8 (Proposition 2.7, [4]). *For $x \neq y \in B$, one has $-S_{xy} > 0$ if and only if there exists a path in G from x to y whose intermediate vertices all lie in B^c .*

Definition 1.9. The *support graph* G_S has vertex set B and an edge between distinct $x, y \in B$ if and only if $S_{xy} \neq 0$. Equivalently, by Lemma 1.8, $\{x, y\}$ is an edge of G or there is a path from x to y with all intermediate vertices in B^c . As a combinatorial graph, G_S depends only on (G, B) .

On a manifold, Wang's weak unique continuation would imply that a Steklov eigenfunction vanishing at a single boundary vertex vanishes identically, since a singleton is open. This fails.

Example 1.1. *Let G be the path $v_1 - v_2 - v_3$ with $B = V$ and $w \equiv 1$. Then Steklov eigenvalues coincide with Laplacian eigenvalues, and*

$$L = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{pmatrix}.$$

The eigenvectors $(1, 1, 1)$, $(1, 0, -1)$ and $(1, -2, 1)$ belong to the eigenvalues 0, 1 and 3 respectively. The second eigenvector vanishes at v_2 but is not identically zero.

Definition 1.10. Let G be connected and $B \subset V$ nonempty. A vertex or an edge belongs to the B -core G^B if it lies on some simple path with both endpoints in B . The B -core is viewed as a subgraph of G .

Interior leaves and hanging blocks—including those attached at an interior vertex, Figure 1—lie outside G^B .

1.2 Vertex Morse functions

A graph has no cotangent bundle, so Morse in the sense of Wang [1] is unavailable. Following Knill [8], a function on vertices is vertex Morse when it is injective on every closed star. The closed star is the combinatorial neighbourhood of a vertex; injectivity there makes every local extremum strict. An earlier index on links of polyhedra appears in Banchoff [9]. Forman's pairing of simplices [10] fails generically and is not used: on the 1-skeleton it requires $|\text{lk}^-(v)| \leq 1$ at every vertex, which a strict local maximum of degree at least two already violates.

The index uses the Euler characteristic of the clique complex; the empty graph has no cliques, so $\chi(\emptyset) = 0$. The characteristic polynomial of S is written $p_S(\lambda, w) = \det(S(w) - \lambda I)$.

Definition 1.11. Let H be a finite simple graph, and let $c_k(H)$ be the number of subgraphs of H isomorphic to K_k for $k \geq 1$. The *Euler characteristic* of the clique complex of H is

$$\chi(H) = \sum_{k \geq 1} (-1)^{k-1} c_k(H).$$

In particular a graph consisting of isolated vertices has characteristic equal to its number of vertices, and a triangle-free graph satisfies $\chi(H) = |V(H)| - |E(H)|$.

Definition 1.12. Let $G = (V, E)$ be a finite simple graph and $f : V \rightarrow \mathbb{R}$. The *link* $\text{lk}(v)$ of a vertex v is the induced subgraph on the neighbours of v . The *closed star* is the induced subgraph on $\{v\} \cup V(\text{lk}(v))$. The function f is a *vertex Morse function* if it is injective on every closed star. The *descending link*

$$\text{lk}^-(v) = \{w \sim v : f(w) < f(v)\}$$

is likewise viewed as an induced subgraph. The *index* is

$$i_f(v) = 1 - \chi(\text{lk}^-(v)).$$

A vertex v is a *critical point* of f if $\chi(\text{lk}^-(v)) \neq 1$. At a local minimum the descending link is empty and $i_f = 1$. A saddle may be critical in this sense without being a local extremum.

Definition 1.13. Let $H = (V_H, E_H)$ be a finite undirected graph and $f : V_H \rightarrow \mathbb{R}$. A vertex $x \in V_H$ is a *local maximum* of f on H if $f(x) \geq f(y)$ for every neighbour y of x in H , and a *strict local maximum* if every such inequality is strict. A *local minimum* and a *strict local minimum* reverse the inequalities. Local maxima and local minima are collectively called *local extrema*.

If $x \in B^c$, harmonicity reads $\sum_{y \sim x} w(x, y)(u(x) - u(y)) = 0$. For $w > 0$ this forbids a strict local extremum at x : all summands would have the same sign. Platforms occur only outside G^B : an interior vertex of degree one, or more generally a hanging block, takes the value of its unique vertex of attachment. Every interior vertex of G^B has degree at least two, Lemma 4.13(iii).

2 Main results

A property of (G, B, w) is said to hold for *generic* positive edge weights if the exceptional set is contained in a finite union of zero sets of functions that are not identically zero. Each such function is either a polynomial on $\mathbb{R}^{|E|}$ or real analytic on the connected open set $\mathbb{R}_+^{|E|}$. A non identically vanishing polynomial has measure-zero zero set with empty interior in $\mathbb{R}_+^{|E|}$, Theorem 3.11; the same holds for a non identically vanishing real-analytic function on $\mathbb{R}_+^{|E|}$.

The following two assertions are due to Chen–Hua–Shi [4]. They will be used as input. The pendant attachment from their inductive construction is recorded in Subsection 3.2, as it is reused below.

Theorem 2.1 (Theorem 1.1, [4]). *Let $G = (V, E)$ be a finite connected graph and let $B \subset V$ be nonempty. For generic $w \in \mathbb{R}_+^{|E|}$,*

- (1) *every Steklov eigenvalue is simple;*
- (2) *every Steklov eigenfunction is nowhere zero on B .*

The constant eigenfunction belonging to $\lambda_0 = 0$ takes the same value at every boundary vertex whenever $|B| \geq 2$, so Theorem 2.1(2) does not give injectivity on B .

Theorem 2.2. *Let $G = (V, E)$ be a finite connected graph and let $B \subset V$ be nonempty. For generic $w \in \mathbb{R}_+^{|E|}$,*

- (1) *every nonconstant Steklov eigenfunction is injective on B ;*
- (2) *the harmonic extension of every nonconstant Steklov eigenfunction is injective on $V(G^B)$ and vanishes at no vertex of V . On each connected component of $G - V(G^B)$ the extension is constant, equal to its value at the unique vertex of attachment.*

If $|B| = 1$, the only eigenfunctions are constants, so (1) is empty. The proofs assume $|B| \geq 2$. If G admits a spanning tree whose leaves all lie in B , then $G^B = G$, and (2) is injectivity on V .

Theorem 2.3. *Let $C \subset \mathbb{R}$ be finite. For generic $w \in \mathbb{R}_+^{|E|}$, every eigenfunction f belonging to a positive Steklov eigenvalue satisfies $\alpha^\top f \neq 0$ whenever $\alpha \in C^B$ is not parallel to $\mathbf{1}$. In particular, taking $C = \{-2, -1, 0, 1, 2\}$,*

- (i) *$2f(i) \neq f(j)$ for all $i \neq j \in B$;*
- (ii) *if $|B| \geq 3$, then $f(i) + f(j) \neq 0$ for all $i \neq j$; together with Theorem 2.2(1), the values of $|f|$ on B are pairwise distinct;*
- (iii) *if $|B| \geq 4$, then $2f(j) \neq f(i) + f(k)$ for all pairwise distinct $i, j, k \in B$.*

If $|B| = 2$, the condition $f(i) + f(j) = 0$ is the relation $f \perp \mathbf{1}$. If $|B| = 3$, a three-term progression forces a coordinate to vanish. Doubling remains an extra condition: $(1, 2, -3)$ is orthogonal to $\mathbf{1}$, nowhere zero, and injective.

On a cycle of length at least 4, assertion (iii) at consecutive vertices is the discrete second difference. Vertex Morse on $G[B]$ follows from injectivity on B , since closed stars in $G[B]$ lie in B . The second difference at a strict local extremum is then automatically nonzero; (iii) concerns all triples. The same three-term condition does not extend to arbitrary triples in G^B : on an equally weighted two-pole path the second difference of the extension vanishes by the resistance interpolation.

Vertex Morse on G requires injectivity on closed stars in G , which may meet B^c . Interior leaves and hanging blocks are constant for every $w > 0$. These are precisely the pieces outside G^B , Figure 1.

Theorem 2.4. *For generic $w \in \mathbb{R}_+^{|E|}$, the harmonic extension u of every nonconstant Steklov eigenfunction is a vertex Morse function on G^B , and*

$$\sum_{v \in V(G^B)} i_u(v) = \chi(G^B).$$

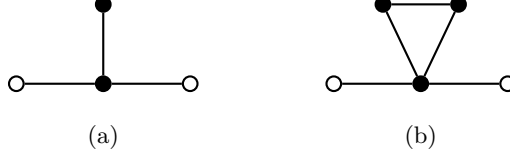


Figure 1: Open vertices lie in B , filled vertices in B^c . Neither graph equals its B -core. (a) An interior leaf. (b) A triangle hanging at an interior vertex of a B - B path. Naive pruning—delete interior degree-one vertices, then delete blocks that meet B in a single vertex—does not remove the triangle in (b). Those two filled vertices still lie on no simple B - B path, hence lie outside G^B .

Injectivity on $V(G^B)$ implies vertex Morse on G^B , since every closed star of G^B lies in $V(G^B)$. Vertex Morse on G^B does not imply injectivity on B : vertex Morse only controls closed stars. On the path $x_0-x_1-x_2-x_3$ the function $(1, 2, 3, 1)$ is vertex Morse, yet $f(x_0) = f(x_3)$. Theorem 2.2(2) gives injectivity on $V(G^B)$. Injectivity on B does not control two neighbours of an interior vertex. On C_4 with B a pair of opposite vertices one has $G^B = G$, yet every spanning tree has a leaf outside B ; see Remark 4.18.

Write $N(C)$ for the set of neighbours in B of a connected component C of $G[B^c]$. The graph G_S is obtained from $G[B]$ by adding a clique on each $N(C)$, Figure 2.

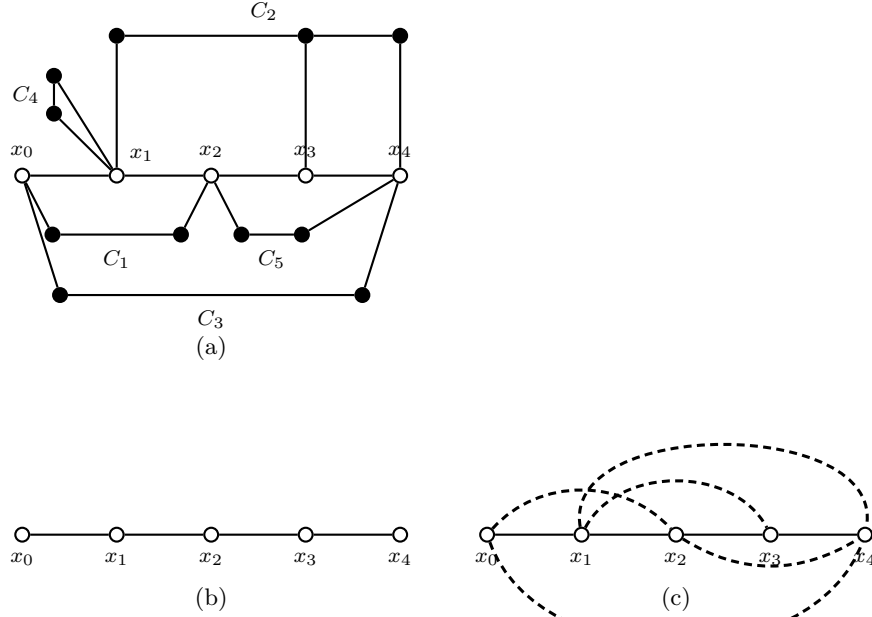


Figure 2: Open vertices lie in B , filled vertices in B^c . (a) The graph G , with five interior components: $N(C_1) = \{x_0, x_2\}$, $N(C_2) = \{x_1, x_3, x_4\}$, $N(C_3) = \{x_0, x_4\}$, $N(C_4) = \{x_1\}$, and $N(C_5) = \{x_2, x_4\}$. (b) The induced graph $G[B]$ is the path $x_0 \cdots x_4$. (c) The support graph G_S : a clique is added on each $N(C)$. Dashed arcs are the clique edges not already in $G[B]$; C_4 contributes none, and $\{x_3, x_4\} \subset N(C_2)$ is already an edge of the path.

Proposition 2.5. *Local extrema of a nonconstant Steklov eigenfunction f on G_S are local extrema of f on $G[B]$, for every $w > 0$. For generic w , f is injective on B ; if G_S is complete, then f has*

exactly two local extrema on G_S , the global maximum and the global minimum. If $G[B]$ is complete, then so is G_S , and the same two vertices are the extrema on both graphs.

Theorem 2.6. *For generic $w \in \mathbb{R}_+^{|E|}$, every nonconstant Steklov eigenfunction is vertex Morse on $G[B]$ and on G_S . The index i_f below is computed on G_S , and*

$$\sum_{v \in B} i_f(v) = \chi(G_S).$$

Moreover:

- (1) *If G_S is a path and $|B| \geq 2$, then for every $w > 0$ the Steklov spectrum is simple. For generic w , an eigenfunction belonging to λ_j , $j \geq 1$, has exactly $j + 1$ local extrema on G_S , including both endpoints. In particular it is strictly monotone for $j = 1$.*
- (2) *If G_S is a cycle and $|B| \geq 3$, an eigenfunction belonging to λ_j , $j \geq 1$, has exactly $2\lceil j/2 \rceil$ local extrema on G_S . In particular λ_1 has exactly two.*
- (3) *If G_S is a tree and $|B| \geq 2$, then for generic w an eigenfunction belonging to λ_1 has a unique sign-changing edge e^* ; on each component of $G_S - e^*$ the values are strictly monotone away from e^* , and the local extrema on G_S are exactly the leaves.*

Theorem 2.7. *Suppose every connected component C of $G[B^c]$ satisfies $|N(C)| \leq 2$. For generic $w \in \mathbb{R}_+^{|E|}$, and for every positive Steklov eigenvalue, the number of nodal domains of the harmonic extension u on V equals the number of nodal domains of $f = u|_B$ on G_S . In particular λ_1 has exactly two nodal domains on V and on $V(G^B)$.*

The same count for λ_1 holds on a star with interior hub, Theorem 4.30, without the two-pole hypothesis. Under the two-pole hypothesis, local extrema of λ_1 on G^B match those of f on G_S , Corollary 4.29. For generic w the extension is nonconstant along every edge of G^B , Theorem 4.14.

The exceptional set of each assertion above has empty interior in $\mathbb{R}_+^{|E|}$.

3 Preliminaries

3.1 First variation formulae

The space of positive edge weights is the open set $\mathbb{R}_+^{|E|}$, whose tangent space may be identified with $\mathbb{R}^E$. Fix $f \in \mathbb{R}^B$ and a tangent vector $h \in \mathbb{R}^E$. For all sufficiently small ε , the weights $w(\varepsilon) = w + \varepsilon h$ remain positive, $L_{[B^c]}(w(\varepsilon))$ remains invertible, and the harmonic extension $u(\varepsilon)$ of f with respect to $w(\varepsilon)$ exists. Write $v = \partial_\varepsilon u|_{\varepsilon=0}$.

Varying a single edge $\{a, b\}$ changes the Laplacian only at the two endpoints:

$$((\partial_{ab}\Delta)u)(x) = \begin{cases} u(a) - u(b), & x = a, \\ u(b) - u(a), & x = b, \\ 0, & \text{otherwise.} \end{cases} \quad (3.1)$$

In matrix form, $\partial_{ab}L = (e_a - e_b)(e_a - e_b)^\top$. For a general tangent vector, $\partial_h L = \sum_{\{a,b\}} h_{ab} (\partial_{ab}L)$.

Lemma 3.1. *Fix $f \in \mathbb{R}^B$. Let $w(\varepsilon) = w + \varepsilon h$, let $u(\varepsilon)$ be the corresponding harmonic extension, and set $v = \partial_\varepsilon u|_{\varepsilon=0}$. Then $v|_B = 0$ and*

$$L_{[B^c]}v_{B^c} = -(\partial_h L_{[B^c]})u_{B^c} - (\partial_h L_{[B^c, B]})f.$$

Equivalently, $\Delta v + (\partial_h \Delta)u = 0$ on the interior.

Proof. For every $x \in B$ and all sufficiently small ε , one has $u(\varepsilon)(x) = f(x)$. Differentiating in ε at $\varepsilon = 0$ yields $v|_B = 0$. Interior harmonicity reads

$$L_{[B^c]}(w(\varepsilon)) u_{B^c}(\varepsilon) + L_{[B^c, B]}(w(\varepsilon)) f = 0.$$

Differentiating in ε and using that f is independent of ε gives the stated equation. $\square$

Lemma 3.2.

$$((D_w \Lambda)(h)f)(x) = (\partial_\nu v)(x) + \sum_{y \sim x} h_{xy} (u(x) - u(y)), \quad x \in B. \quad (3.2)$$

Proof. For $x \in B$,

$$(\Lambda_{w(\varepsilon)} f)(x) = \sum_{y \sim x} w(\varepsilon)_{xy} (u(\varepsilon)(x) - u(\varepsilon)(y)).$$

Differentiating the product on each edge and evaluating at $\varepsilon = 0$ yields

$$\partial_\varepsilon|_{\varepsilon=0} (\Lambda_{w(\varepsilon)} f)(x) = \sum_{y \sim x} h_{xy} (u(x) - u(y)) + \sum_{y \sim x} w_{xy} (v(x) - v(y)).$$

Lemma 3.1 gives $v(x) = 0$, so the second sum equals $(\partial_\nu v)(x)$. $\square$

Differentiating the Schur complement (1.7) directly produces the same formula. Since S is a rational function of w and is regular for $w > 0$, the map $w \mapsto S(w)$ is smooth on $\mathbb{R}_+^{|E|}$.

Lemma 3.3. *If $\{a, b\} \subset B^c$, then*

$$D_{ab} S = (p_a - p_b)(p_a - p_b)^\top,$$

and therefore $((D_w \Lambda)(\partial_{ab})f) = (u(a) - u(b))(p_a - p_b)$.

Proof. The edge $\{a, b\}$ appears only in the interior block, so $\partial_{ab} L_{[B^c]} = (e_a - e_b)(e_a - e_b)^\top$ and the remaining blocks are stationary. Differentiating the inverse via $\partial(A^{-1}) = -A^{-1}(\partial A)A^{-1}$ gives

$$D_{ab} S = L_{[B, B^c]} L_{[B^c]}^{-1} (e_a - e_b)(e_a - e_b)^\top L_{[B^c]}^{-1} L_{[B^c, B]} = (p_a - p_b)(p_a - p_b)^\top,$$

where the last step uses $L_{[B, B^c]} L_{[B^c]}^{-1} e_u = -p_u$. Applying the resulting rank-one matrix to f and invoking (1.10) yields the second identity. $\square$

Lemma 3.4. *If $\{x, y\} \subset B$, then $v \equiv 0$ and*

$$D_{xy} S = (e_x - e_y)(e_x - e_y)^\top,$$

so that $((D_w \Lambda)(\partial_{xy})f) = (f(x) - f(y))(e_x - e_y)$.

Proof. The edge $\{x, y\}$ appears only in $L_{[B]}$. Lemma 3.1 therefore gives $L_{[B^c]} v_{B^c} = 0$, hence $v \equiv 0$. Differentiating S leaves only $\partial_{xy} L_{[B]} = (e_x - e_y)(e_x - e_y)^\top$. $\square$

Lemma 3.5. *If $x \in B$, $z \in B^c$ and $\{x, z\} \in E$, then*

$$D_{xz} S = (e_x - p_z)(e_x - p_z)^\top,$$

and therefore $((D_w \Lambda)(\partial_{xz})f) = (f(x) - u(z))(e_x - p_z)$.

Proof. The relevant block derivatives are

$$\partial_{xz}L_{[B]} = e_x e_x^\top, \quad \partial_{xz}L_{[B^c]} = e_z e_z^\top, \quad \partial_{xz}L_{[B, B^c]} = -e_x e_z^\top.$$

Differentiating $S = L_{[B]} - L_{[B, B^c]}L_{[B^c]}^{-1}L_{[B^c, B]}$ and using $p_z = -L_{[B, B^c]}L_{[B^c]}^{-1}e_z$ together with symmetry of $L_{[B^c]}^{-1}$ produces four terms which combine as

$$e_x e_x^\top - e_x p_z^\top - p_z e_x^\top + p_z p_z^\top = (e_x - p_z)(e_x - p_z)^\top.$$

The identity $u(z) = p_z^\top f$ from (1.10) gives the second statement. $\square$

Assign to each vertex a vector in $\mathbb{R}^B$ by

$$\psi(v) = \begin{cases} e_v, & v \in B, \\ p_v, & v \in B^c. \end{cases} \quad (3.3)$$

Then $\mathbf{1}^\top \psi(v) = 1$ for every v , so $\psi(a) - \psi(b) \perp \mathbf{1}$. The harmonic extension satisfies $u(v) = \psi(v)^\top f$.

Lemma 3.6. *For every edge $\{a, b\} \in E$,*

$$D_{ab}S = (\psi(a) - \psi(b))(\psi(a) - \psi(b))^\top$$

and

$$((D_w \Lambda)(\partial_{ab})f) = (u(a) - u(b))(\psi(a) - \psi(b)).$$

Proof. The three possible locations of the endpoints $\{a, b\}$ are covered by Lemmas 3.4, 3.3 and 3.5 respectively. In each case $u(v) = \psi(v)^\top f$, which identifies the two displayed formulae. $\square$

Lemma 3.7. *Let λ be a simple eigenvalue of S , with unit eigenvector f and harmonic extension u . Then, for every edge $\{a, b\}$,*

$$\frac{\partial \lambda}{\partial w_{ab}} = (u(a) - u(b))^2.$$

In particular, increasing any single edge weight cannot decrease λ . Moreover S is homogeneous of degree one in the edge weights, and

$$\sum_{\{a, b\} \in E} w_{ab} \frac{\partial \lambda}{\partial w_{ab}} = \sum_{\{a, b\} \in E} w_{ab} (u(a) - u(b))^2 = u^\top L u = \lambda.$$

Proof. Let $w(\varepsilon)$ vary along a line and write $S(\varepsilon)$ for the corresponding Steklov matrix. Since λ is simple and $S(\varepsilon)$ is real symmetric and analytic in ε , there exist differentiable branches $\lambda(\varepsilon)$ and $f(\varepsilon)$ with $S(\varepsilon)f(\varepsilon) = \lambda(\varepsilon)f(\varepsilon)$ and $f(\varepsilon)^\top f(\varepsilon) = 1$. Differentiating at $\varepsilon = 0$ and left-multiplying by $f^\top$ yields $\partial \lambda = f^\top (\partial S) f$. Taking the variation along a single edge and applying Lemma 3.6 gives

$$f^\top (D_{ab}S) f = (u(a) - u(b))^2.$$

If all edge weights are scaled by $t > 0$, then $S(tw) = tS(w)$, so $\lambda(tw) = t\lambda(w)$. Differentiating at $t = 1$ produces the Euler identity. The last equality is Lemma 1.6. $\square$

3.2 Generic simplicity and nonvanishing

The following analytic and algebraic facts are standard. Statements 3.8–3.11 will be used below. Lemmas 3.12–3.14 are taken from Chen–Hua–Shi [4].

Theorem 3.8 (Theorem 2.1, [5]). *Let $A(p)$ be a real symmetric matrix whose entries depend analytically on a parameter p . If λ is a simple eigenvalue of $A(p^*)$ with eigenvector x , then there exist analytic branches $\lambda(p)$ and $x(p)$ near p^* such that $A(p)x(p) = \lambda(p)x(p)$, with $\lambda(p^*) = \lambda$ and $x(p^*) = x$.*

Theorem 3.9. *Let $A = \begin{pmatrix} B & y \\ y^\top & a \end{pmatrix}$ be an $(n+1) \times (n+1)$ Hermitian matrix, with B of size n . Then*

$$\lambda_1(A) \leq \lambda_1(B) \leq \lambda_2(A) \leq \cdots \leq \lambda_n(B) \leq \lambda_{n+1}(A).$$

Equality $\lambda_i(A) = \lambda_i(B)$ holds if and only if there exists an eigenvector z of B belonging to that eigenvalue with $y^\top z = 0$; in that case $(z^\top, 0)^\top$ is an eigenvector of A .

Proposition 3.10 (Proposition 2.5, [4]). *A Hermitian matrix A admits an eigenvector with a vanishing coordinate if and only if A shares an eigenvalue with some principal submatrix of size $n-1$.*

Theorem 3.11 (2.6.5, [6]). *The zero set of a non identically vanishing polynomial $P : \mathbb{R}^n \rightarrow \mathbb{R}$ has Lebesgue measure zero and Hausdorff dimension at most $n-1$.*

Lemma 3.12 (Proposition 2.9 and Corollary 2.11, [4]). *Let P be a path whose boundary B_P consists of the two endpoints, and set $\alpha = (\sum_{e \in P} w(e)^{-1})^{-1}$. For every positive edge weight, the Steklov matrix is*

$$S = \alpha \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

The spectrum is $\{0, 2\alpha\}$, hence simple. A nonconstant eigenfunction takes the boundary values $(c, -c)$ with $c \neq 0$, and is therefore nowhere zero on B_P . Along the path $x_0, \dots, x_m$ the harmonic extension is the affine interpolation

$$h(x_j) = h(x_0) + \theta_j(h(x_m) - h(x_0)), \quad \theta_j = \frac{\sum_{t=1}^j w(x_{t-1}, x_t)^{-1}}{\sum_{t=1}^m w(x_{t-1}, x_t)^{-1}} \in (0, 1). \quad (3.4)$$

If $h(x_0) \neq h(x_m)$, an interior value vanishes if and only if some θ_j equals $h(x_0)/(h(x_0) - h(x_m))$. The matrix S depends on the path only through $\sum w(e)^{-1}$. Keeping this sum fixed and varying the summands avoids any finite set of exceptional ratios, so the weights may be chosen so that h vanishes at no interior vertex.

Lemma 3.13 (Proposition 2.9, [4]). *Let v be an interior vertex with exactly two neighbours u_1, u_2 , and assume $\{u_1, u_2\} \notin E$. Replacing the path u_1-v-u_2 by a single edge of weight*

$$w'(u_1, u_2) = \frac{w(u_1, v)w(v, u_2)}{w(u_1, v) + w(v, u_2)}$$

does not change the Steklov matrix.

Lemma 3.14 (Lemma 2.12, [4]). *Let S be the Steklov matrix of (G, B, w) . Attach a pendant edge of weight $\tau > 0$ at a vertex u , and include the new leaf b in the boundary. The new Steklov matrix is*

$$\widehat{S} = \begin{cases} \begin{pmatrix} S & 0 \\ 0 & 0 \end{pmatrix} + \tau \begin{pmatrix} e_u e_u^\top & -e_u \\ -e_u^\top & 1 \end{pmatrix}, & u \in B, \\ \begin{pmatrix} S & 0 \\ 0 & 0 \end{pmatrix} + \frac{\tau}{1 + \tau\alpha} \begin{pmatrix} p_u p_u^\top & -p_u \\ -p_u^\top & 1 \end{pmatrix}, & u \in B^c, \end{cases} \quad (3.5)$$

where $\alpha = e_u^\top L_{[B^c]}^{-1} e_u > 0$ and $p_u = -L_{[B, B^c]} L_{[B^c]}^{-1} e_u$ (with e_u viewed as a basis vector of $\mathbb{R}^{|B^c|}$ in the second case). Moreover $\mathbf{1}_B^\top p_u = 1$.

Throughout the remainder of the paper the following notation will be used. After attaching a pendant edge as in Lemma 3.14, write

$$\widehat{S}(\tau) = S_0 + \kappa(\tau) M(v), \quad S_0 = \begin{pmatrix} S_P & 0 \\ 0 & 0 \end{pmatrix}, \quad M(v) = \begin{pmatrix} v v^\top & -v \\ -v^\top & 1 \end{pmatrix}, \quad (3.6)$$

where S_P is the Steklov matrix of the original graph, $v = e_u$ and $\kappa = \tau$ if $u \in B$, while $v = p_u$ and $\kappa = \tau/(1 + \tau\alpha)$ if $u \in B^c$. In both cases $\mathbf{1}^\top v = 1$ and $M(v)\mathbf{1}_{n+1} = 0$, so $\widehat{S}(\tau)\mathbf{1}_{n+1} = 0$. As $\tau \rightarrow 0^+$ one has $\kappa \rightarrow 0$, hence $\widehat{S}(\tau) \rightarrow S_0$. If $\ker S_P = \text{span}\{\mathbf{1}_n\}$, then

$$\ker S_0 = \text{span}\left\{\begin{pmatrix} \mathbf{1}_n \\ 0 \end{pmatrix}, \begin{pmatrix} 0_n \\ 1 \end{pmatrix}\right\},$$

and the remaining eigenvalues of S_0 are the positive eigenvalues of S_P . Restricting to $\mathbf{1}_{n+1}^\perp$ therefore makes the eigenvalue 0 of S_0 simple, with

$$\ker S_0 \cap \mathbf{1}_{n+1}^\perp = \text{span}\{(-\mathbf{1}_n^\top, n)^\top\}.$$

Lemma 3.15. *In the notation of (3.6), suppose $\widehat{S}(\tau)x = \lambda x$ with $x = (x_P^\top, x_b)^\top$ and $x_b = 0$. Then $v^\top x_P = 0$ and $S_P x_P = \lambda x_P$. If $u \in B$ this forces $x_P(u) = 0$; if $u \in B^c$ the old harmonic extension vanishes at u .*

Proof. The identity (3.6) expands as

$$\begin{aligned} S_P x_P + \kappa(v^\top x_P - x_b)v &= \lambda x_P, \\ \kappa(x_b - v^\top x_P) &= \lambda x_b. \end{aligned}$$

Substituting $x_b = 0$ yields $v^\top x_P = 0$ and $S_P x_P = \lambda x_P$. Then $v = e_u$ or $v = p_u$ according as $u \in B$ or $u \in B^c$. $\square$

The proofs of Theorems 2.1 and 2.2 follow a common pattern. One first constructs positive weights on a spanning tree for which the relevant discriminant polynomial is nonzero, then sets the weights of all non-tree edges to zero. The resulting point of $\mathbb{R}^{|E|}$ is not a pole of the discriminant, since the interior block of the tree remains invertible, and the polynomial of the original graph does not vanish there, hence is not identically zero.

If a leaf of the spanning tree T does not lie in B , harmonicity forces that leaf to take the same value as its unique neighbour. Deleting the leaf does not change the Steklov matrix, since the corresponding pendant weight does not appear in S . The discriminant polynomials of G therefore

coincide with those of the pruned graph as functions of the remaining weights, and it may be assumed that every leaf of T lies in B .

Theorem 2.1 is [4, Theorem 1.1]. The inductive construction on a spanning tree, via Lemmas 3.12–3.14 and (3.6), is reused below; the argument is not repeated.

Write $p_S(\lambda, w) = \det(S(w) - \lambda I)$. Set

$$\text{Discr}_s(w) = \text{Res}(p_S(\cdot, w), \partial_\lambda p_S(\cdot, w)) = \frac{P_s(w)}{(\det L_{[B^c]})^N}.$$

Then $\text{Discr}_s(w) = 0$ if and only if the spectrum is not simple. For each $i \in B$, let p_i be the characteristic polynomial of the principal submatrix of S obtained by deleting the i -th coordinate, and set

$$\text{Discr}_0(w) = \text{Res}\left(p_S(\lambda, w), \prod_{i \in B} p_i(\lambda, w)\right) = \frac{P_0(w)}{(\det L_{[B^c]})^N}.$$

Then $\text{Discr}_0(w) = 0$ if and only if some eigenvector vanishes at some boundary vertex. Theorem 2.1 asserts that neither P_s nor P_0 is identically zero.

Theorem 3.16. *Let G be connected. For generic edge weights, the harmonic extension of every nonconstant Steklov eigenfunction vanishes at no vertex of V .*

Proof. By Theorem 2.1(2), $f(b) \neq 0$ for every $b \in B$. If $B^c = \emptyset$ there is nothing to prove.

On each connected component C of $G - V(G^B)$ there is a unique vertex of attachment x_0 , and

$$(\Delta h)|_C = 0, \quad h|_{\partial C} = h(x_0),$$

so $h|_C \equiv h(x_0)$. Thus it is enough that $h(x) \neq 0$ for all $x \in V(G^B)$.

Let T be a tree whose leaves all lie in B and $|B| \geq 2$. Let $\mathcal{P}^{\text{nz}}(n)$ assert the existence of a subtree $T^{(n)}$ with $|B^{(n)}| = n$ and a positive weight such that the spectrum is simple and $H(x) \neq 0$ for every $x \in V(T^{(n)})$ and every nonconstant harmonic extension H .

If $n = 2$, let $P \subset T$ have endpoints in B . Lemma 3.12 gives

$$S = \alpha \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad f = (c, -c), \quad c \neq 0,$$

and (3.4) yields $H(x_j) \neq 0$ after a perturbation of the weights. Thus $\mathcal{P}^{\text{nz}}(2)$ holds.

Assume $\mathcal{P}^{\text{nz}}(n)$. Choose $b \in B \setminus B^{(n)}$ so that P_{ub} meets B only at its endpoints, and let $u = P_{ub} \cap T^{(n)}$. Lemmas 3.13 and 3.14 contract P_{ub} to a pendant edge of weight $\tau > 0$:

$$\hat{S}(\tau) = S_0 + \kappa(\tau) M(v), \quad x(\tau) = \begin{pmatrix} x_P(\tau) \\ x_b(\tau) \end{pmatrix}.$$

For $i \geq 2$, Theorem 3.8 supplies

$$x_P(0) = x_P, \quad x_b(0) = 0,$$

with $x_P(y) \neq 0$ for all $y \in V(T^{(n)})$ by the inductive hypothesis. Continuity gives $x_P(\tau)(y) \neq 0$ for small τ . Lemma 3.15 yields that $x_b(\tau) = 0$ would force $x_P(\tau)(u) = 0$ or $p_u^\top x_P(\tau) = 0$, a contradiction. Hence $x_b(\tau) \neq 0$.

For $\hat{\lambda}_1$,

$$\ker S_0 \cap \mathbf{1}_{n+1}^\perp = \text{span}\{(-\mathbf{1}_n^\top, n)^\top\},$$

so a unit eigenvector in $\mathbf{1}^\perp$ satisfies

$$x(\tau) \rightarrow c \begin{pmatrix} -\mathbf{1}_n \\ n \end{pmatrix}, \quad c = \frac{1}{\sqrt{n(n+1)}}.$$

Every coordinate of the limit is nonzero, hence $x(\tau)$ has no zero coordinate for small τ . Values at old interior vertices tend to $-c \neq 0$.

On P_{ub} , (3.4) gives $H(x_j) = 0$ if and only if $\theta_j = H(u)/(H(u) - H(b))$ (when $H(u) \neq H(b)$). Keeping $\sum w^{-1} = 1/\tau$ fixed, the finitely many exceptional $\theta_j \in (0, 1)$ may be avoided. This is $\mathcal{P}^{\text{nz}}(n+1)$.

Thus on any tree with leaves in B there is a positive weight at which every nonconstant extension is nowhere zero.

Now fix $x \in V(G^B) \cap B^c$. Take a path P through x with both endpoints in B . Attach each remaining $b \in B$ along a path meeting the current tree at a single vertex. The resulting tree T_* satisfies $V(T_*) \supset B \cup \{x\}$ and has all leaves in B . The preceding construction supplies w_* on T_* with $H(x) \neq 0$ for every nonconstant extension. Let T_0 be a spanning tree of G containing T_* , and set

$$w^{(\varepsilon)}(e) = \begin{cases} w_*(e), & e \in E(T_*), \\ 1, & e \in E(T_0) \setminus E(T_*), \\ \varepsilon, & e \notin E(T_0). \end{cases}$$

Then $S(w^{(\varepsilon)}) \rightarrow S(T_*, w_*)$ and $\det L_{[B^c]}(w^{(0)}) > 0$. For small $\varepsilon > 0$, every nonconstant extension satisfies $u(x) \neq 0$.

Fix $x \in B^c$. By (1.10), $h_f(x) = p_x^\top f$. Let $p_S(\lambda, w) = \det(S(w) - \lambda I)$ and

$$\mu_x(\lambda, w) = p_x(w)^\top \text{adj}(S(w) - \lambda I) p_x(w), \quad \text{Discr}_x(w) = \text{Res}(p_S(\cdot, w), \mu_x(\cdot, w)) = \frac{P_x(w)}{(\det L_{[B^c]})^N}.$$

If $Sf = \lambda f$ with λ simple, then $\text{adj}(S - \lambda I) = c_\lambda f f^\top$ for some $c_\lambda \neq 0$, hence

$$\mu_x(\lambda) = 0 \iff p_x^\top f = 0 \iff h_f(x) = 0.$$

If some eigenvalue has multiplicity at least two, then $\text{adj}(S - \lambda I) = 0$, so $\text{Discr}_x(w) = 0$. The constant eigenfunction satisfies $p_x^\top \mathbf{1} = 1$, hence $\mu_x(0) \neq 0$.

If $x \in V(G^B)$, the open set just constructed shows $P_x \neq 0$. If $x \notin V(G^B)$, then $h(x) = h(x_0)$ at the attachment, so $\{P_x = 0\}$ is contained in the corresponding zero set for x_0 . There are finitely many vertices, and the union of zero sets has measure zero by Theorem 3.11. $\square$

4 Proofs

4.1 Injectivity on the boundary

The constant eigenfunction is constant on B . Throughout the proof of Theorem 2.2 only positive eigenvalues are considered, and $|B| \geq 2$.

Lemma 4.1. *A real symmetric matrix A admits an eigenvector f with $f(i) = f(j)$ if and only if A shares an eigenvalue with its compression to the hyperplane $(e_i - e_j)^\perp$.*

Proof. Let $q_n = (e_i - e_j)/\sqrt{2}$ and let $Q = (U, q_n)$ be orthogonal, so that the columns of U span $(e_i - e_j)^\perp$. Then

$$f(i) = f(j) \iff q_n^\top f = 0 \iff (Q^\top f)_n = 0.$$

Write $\tilde{A} = Q^\top A Q$, so

$$\tilde{A} = \begin{pmatrix} B & y \\ y^\top & a \end{pmatrix}, \quad B = U^\top A U.$$

The block B is a principal submatrix of $\tilde{A}$. By Proposition 3.10,

$$\tilde{A} \tilde{f} = \lambda \tilde{f} \text{ with } (\tilde{f})_n = 0 \iff \tilde{A} \text{ and } B \text{ share an eigenvalue.}$$

Since $\tilde{A}$ is orthogonally similar to A , this is the claim. $\square$

The constant vector lies in every hyperplane $(e_i - e_j)^\perp$. On a connected graph it spans $\ker S$, so 0 is always a common eigenvalue of S and of each compression. The corresponding resultant would vanish identically if this eigenvalue were not removed.

Lemma 4.2. *Assume that 0 is a simple eigenvalue of S with $\ker S = \text{span}\{\mathbf{1}\}$. Then S admits a nonconstant eigenvector f with $f(i) = f(j)$ if and only if S and the compression $B_{ij} = U_{ij}^\top S U_{ij}$ share a nonzero eigenvalue, where the columns of U_{ij} form an orthonormal basis of $(e_i - e_j)^\perp$.*

Proof. Let $Q = (U_{ij}, q_n)$ be as in Lemma 4.1, and write

$$\tilde{S} = Q^\top S Q = \begin{pmatrix} B_{ij} & y \\ y^\top & a \end{pmatrix}. \quad (4.1)$$

A vector f satisfies $f(i) = f(j)$ if and only if $f = U_{ij}\xi$ for some $\xi \in \mathbb{R}^{n-1}$. In that case

$$Sf = U_{ij}(B_{ij}\xi) + (y^\top \xi) q_n. \quad (4.2)$$

If moreover $B_{ij}\xi = \lambda\xi$, then $Sf = \lambda f + (y^\top \xi) q_n$.

Since $\mathbf{1} \in (e_i - e_j)^\perp$, one may write $\mathbf{1} = U_{ij}\xi_1$. Then $B_{ij}\xi_1 = 0$, so 0 is always a common eigenvalue. If $B_{ij}\xi = 0$ and $f = U_{ij}\xi$, then $f^\top S f = \xi^\top B_{ij}\xi = 0$, hence $Sf = 0$ by positive semidefiniteness, so $f \in \text{span}\{\mathbf{1}\}$. Thus $\ker B_{ij} = \text{span}\{\xi_1\}$ and 0 remains simple for B_{ij} .

By Lemma 4.1, S admits an eigenvector with equal i -th and j -th coordinates if and only if S and B_{ij} share an eigenvalue. The only common eigenvalue corresponding to a constant eigenvector is 0. A nonconstant eigenvector with $f(i) = f(j)$ therefore exists if and only if a positive common eigenvalue exists.

In one direction, if $Sf = \lambda f$ with $f(i) = f(j)$ and $\lambda > 0$, then $f = U_{ij}\xi$ and $B_{ij}\xi = \lambda\xi$. Conversely, if $\lambda > 0$ is a common eigenvalue, Cauchy interlacing applied to (4.1) forces at least one equality in the interlacing chain. The equality case of Theorem 3.9 produces z with $B_{ij}z = \lambda z$ and $y^\top z = 0$, so that $f = Q(z^\top, 0)^\top$ is an eigenvector of S with $f(i) = f(j)$ and $\lambda > 0$. $\square$

Write $p_S(\lambda, w) = \det(S(w) - \lambda I)$ and $\tilde{p}_S = p_S/\lambda$. Likewise set $\tilde{p}_{ij}(\lambda, w) = \det(B_{ij}(w) - \lambda I)/\lambda$, and define

$$\text{Discr}_{\text{inj}}(w) = \prod_{i < j} \text{Res}(\tilde{p}_S(\cdot, w), \tilde{p}_{ij}(\cdot, w)). \quad (4.3)$$

For each pair $i < j$, the corresponding resultant vanishes if and only if S and B_{ij} share a nonzero eigenvalue. As before,

$$\text{Discr}_{\text{inj}}(w) = \frac{P_{\text{inj}}(w)}{(\det L_{[B^c]})^N},$$

and $P_{\text{inj}} \neq 0$ is equivalent to generic injectivity of nonconstant eigenfunctions on B .

Let S^+ denote the inverse of the restriction of S to $\mathbf{1}^\perp$. For $v \in \mathbb{R}^B$ write $\bar{v} = n^{-1} \mathbf{1}^\top v$. The equation $S\phi = v - \bar{v} \mathbf{1}$ with $\phi \perp \mathbf{1}$ has a unique solution, denoted $\phi = S^+ v$. Equivalently,

$$S^+ = (S + \mathbf{1} \mathbf{1}^\top / n)^{-1} - \mathbf{1} \mathbf{1}^\top / n.$$

The entries of S^+ are rational functions of the edge weights.

Lemma 4.3. *Let T be a tree whose boundary B contains every leaf of T , with $|B| = n \geq 2$, and let S be the corresponding Steklov matrix.*

- (a) *If $u \in B$ and $v = e_u$, then for almost every positive edge weight the function $\phi = S^+ v$ takes pairwise distinct values on B .*
- (b) *If $u \in B^c$, then for almost every positive edge weight the vector p_u is nonconstant and $S^+ p_u$ takes pairwise distinct values on B .*

Proof. Write $\phi = S^+ v$, so that $S\phi = v - \bar{v} \mathbf{1}$ and $\phi \perp \mathbf{1}$. Let h be the harmonic extension of ϕ . Then $(Lh)|_B = v - \bar{v} \mathbf{1}$ and $(Lh)|_{B^c} = 0$.

Any edge $e = \{a, b\}$ of T splits T into subtrees $T_1 \sqcup T_2$ with $a \in T_1$ and $b \in T_2$. Since every leaf lies in B , both T_1 and T_2 meet B , and $|B \cap T_1| \in \{1, \dots, n-1\}$. Summing Lh over T_1 , interior edges cancel and the unique crossing edge contributes $w(e)(h(a) - h(b))$. The same sum equals

$$c_e := \sum_{x \in T_1 \cap B} (v_x - \bar{v}),$$

so

$$w(e)(h(a) - h(b)) = c_e. \quad (4.4)$$

(The identity $L\mathbf{1} = 0$ implies that the corresponding sum over T_2 equals $-c_e$.)

For $k \neq l \in B$, let $k = x_0, \dots, x_r = l$ be the unique path. Summing (4.4) along the path yields

$$\phi(k) - \phi(l) = \sum_{j=1}^r \frac{c_{e_j}}{w(e_j)}. \quad (4.5)$$

The set $\{\phi(k) = \phi(l)\}$ is the zero set of this rational function. It remains to show that the function is not identically zero.

- (a) Take $v = e_u$ with $u \in B$. Then $\bar{v} = n^{-1}$ and, writing $m = |B \cap T_1|$,

$$c_e = 1_{\{u \in T_1\}} - \frac{m}{n}.$$

Since $m/n \in (0, 1)$, the constant c_e is nonzero whether or not $u \in T_1$, and is independent of the edge weights. For any $k \neq l$, every edge of the path has $c_{e_j} \neq 0$. Fixing all but one of those weights, (4.5) is a non identically vanishing rational function of the remaining weight. Thus $\phi(k) - \phi(l) \neq 0$.

(b) First, p_u is not identically constant. By (1.10), $p_u(x) = h_{\delta_x}(u)$ and $\mathbf{1}^\top p_u = 1$. If $p_u \equiv n^{-1} \mathbf{1}$, take a leaf $b \in B$ with unique neighbour t and incident weight τ . As $\tau \rightarrow 0^+$, the edge $\{b, t\}$ disappears from the Laplacian. The remaining tree, with boundary $B \setminus \{b\}$ and vanishing boundary values, has unique harmonic extension identically zero, so $p_u(b) \rightarrow 0$, contradicting $p_u(b) \equiv 1/n$.

Now fix $k \neq l \in B$ and write $F_{kl}(w) = (e_k - e_l)^\top S(w)^+ p_u(w)$. Let $e = \{a, b\}$ be an edge of $\text{path}(k, l)$, splitting T into $T_k \sqcup T_l$ with $k \in T_k$ and $l \in T_l$. Assume $u \in T_l$. Then

$$c_e = \sum_{x \in T_k \cap B} (p_u(x) - n^{-1}) = \sigma - \frac{m}{n},$$

where $m = |B \cap T_k|$ and σ is the value at u of the harmonic extension of $1_{B \cap T_k}$. Fixing the remaining weights and sending $w(e) \rightarrow 0^+$, the Dirichlet problem on T_l tends to the problem with vanishing boundary values, whose solution is identically zero. Thus $\sigma \rightarrow 0$ and $c_e \rightarrow -m/n \neq 0$. For sufficiently small $w(e)$ one has $|c_e| > m/(2n)$.

A discrete harmonic function attains its maximum and minimum on the boundary. Consequently $0 \leq p_u \leq 1$, and $|c_{e'}| \leq n$ for every other edge e' of the path, uniformly in $w(e)$. Therefore

$$F_{kl} = \frac{c_e}{w(e)} + O(1)$$

as $w(e) \rightarrow 0^+$, so $|F_{kl}| \rightarrow \infty$. In particular $F_{kl} \neq 0$.

Each F_{kl} is a rational function P_{kl}/Q_{kl} . The denominator arises from the two inversions $L_{[B^c]}^{-1}$ and $(S + \mathbf{1}\mathbf{1}^\top/n)^{-1}$. For $w > 0$ both determinants are strictly positive, so F_{kl} has no poles on $\mathbb{R}_+^{[E]}$ and $F_{kl} = 0$ if and only if $P_{kl} = 0$. There are finitely many pairs $k \neq l$, so the union of the zero sets has measure zero. $\square$

The first positive eigenvalue of a pendant attachment cannot be analysed at order zero. The limiting eigenvector is constant on the old boundary. The next lemma records the first-order correction.

Lemma 4.4. *The matrix $\hat{S}$ annihilates $\mathbf{1}_{n+1}$ for every κ , and may therefore be restricted to $\mathbf{1}^\perp$. On this hyperplane the eigenvalue 0 of S_0 is simple. Theorem 3.8 supplies analytic branches $\hat{\lambda}_1(\kappa)$ and a unit eigenvector $x(\kappa) = (y(\kappa), z(\kappa))$ with $z \geq 0$, satisfying $\hat{\lambda}_1 = \kappa\mu$, $\mu(0) = (n+1)/n$, and*

$$y = -c\mathbf{1} + \kappa c(n+1)S_P^+v + O(\kappa^2), \quad z = nc + O(\kappa^2),$$

where $c = 1/\sqrt{n(n+1)}$ and $v = e_u$ or $v = p_u$. If S_P^+v is injective on $B^{(r)}$, then for all sufficiently small κ the vector y is injective and $y_k \neq z$ for every k .

Proof. Restriction to $\mathbf{1}_{n+1}^\perp$ and the zeroth-order identification

$$y(0) = -c\mathbf{1}, \quad z(0) = nc$$

follow from $\ker S_0 \cap \mathbf{1}_{n+1}^\perp = \text{span}\{(-\mathbf{1}_n^\top, n)^\top\}$. Analyticity gives $\hat{\lambda}_1(\kappa) = \kappa\mu(\kappa)$.

The eigenvalue equations are

$$S_P y + \kappa(v^\top y - z)v = \lambda y, \tag{4.6}$$

$$\kappa(z - v^\top y) = \lambda z. \tag{4.7}$$

Substitute $y = -c\mathbf{1} + \kappa y^{(1)} + O(\kappa^2)$ and $z = nc + \kappa z^{(1)} + O(\kappa^2)$ into the orthogonality condition $z = -\mathbf{1}^\top y$. Comparing first-order terms yields $z^{(1)} = -\mathbf{1}^\top y^{(1)}$. Differentiating the unit-length constraint $y^\top y + z^2 = 1$ at $\kappa = 0$ and substituting the same expansions gives $\mathbf{1}^\top y^{(1)} = 0$, hence $z^{(1)} = 0$. Thus $z = nc + O(\kappa^2)$.

From (4.7), $v^\top y = z(1 - \mu)$. Passing to the limit $\kappa \rightarrow 0$ and using $\mathbf{1}^\top v = 1$ yields $\mu(0) = (n+1)/n$. Substituting $v^\top y - z = -\mu z$ into (4.6) and comparing first-order terms, using $S_P \mathbf{1} = 0$, produces

$$S_P y^{(1)} = c(n+1)(v - n^{-1}\mathbf{1}).$$

The right-hand side is orthogonal to $\mathbf{1}$, hence lies in the range of S_P . Together with $\mathbf{1}^\top y^{(1)} = 0$ this uniquely determines $y^{(1)} = c(n+1)S_P^+v$.

Consequently

$$y_k - y_l = \kappa c(n+1)((S_P^+ v)_k - (S_P^+ v)_l) + O(\kappa^2).$$

If $S_P^+ v$ is injective on $B^{(r)}$, the coefficient of κ is nonzero for every $k \neq l$, so y is injective for small κ . Moreover $z - y_k = (n+1)c + O(\kappa)$ is nonzero for small κ . $\square$

Proof of Theorem 2.2(1). By (4.3) and Lemma 1.3,

$$\text{Discr}_{\text{inj}}(w) = \frac{P_{\text{inj}}(w)}{(\det L_{[B^c]})^N},$$

and $\text{Discr}_{\text{inj}}(w) = 0$ if and only if $P_{\text{inj}}(w) = 0$ whenever $w > 0$. It suffices to show $P_{\text{inj}} \neq 0$.

Let $T = (V, E_T)$ be a spanning tree, and let w_T be a positive weight on T . Define

$$w^*(e) = \begin{cases} w_T(e), & e \in E_T, \\ 0, & e \notin E_T. \end{cases}$$

Then $L(G, w^*) = L(T, w_T)$, hence $S(G, w^*) = S(T, w_T)$ and $B_{ij}(w^*) = B_{ij}^T(w_T)$. In particular

$$\text{Discr}_{\text{inj}}(w^*) = \text{Discr}_{\text{inj}}^T(w_T).$$

If $P_{\text{inj}}^T(w_T) \neq 0$, then $P_{\text{inj}}(w^*) \neq 0$. Leaves of T not in B may be deleted without changing S . Write $|B| = N$.

Let $\mathcal{P}(n)$ assert the existence of a subtree $T^{(n)} \subset T$, a set $B^{(n)} \subset B$ with $|B^{(n)}| = n$, and a positive weight $w^{(n)}$ on $T^{(n)}$ such that Theorem 2.1 holds, every nonconstant Steklov eigenfunction of $(T^{(n)}, B^{(n)}, w^{(n)})$ is injective on $B^{(n)}$, and every nonconstant harmonic extension is nowhere zero on $V(T^{(n)})$.

If $n = 2$, let $P \subset T$ be a path with both endpoints in B . Lemma 3.12 gives

$$S = \alpha \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \quad f = (c, -c), \quad c \neq 0,$$

so f is injective on the two-point boundary, and (3.4) gives a weight with no interior zero.

Assume $\mathcal{P}(n)$ for some $2 \leq n < N$. Choose $b \in B \setminus B^{(n)}$ so that the unique path P_{ub} in T from b to $T^{(n)}$ meets B only at its endpoints, and let u be the intersection with $T^{(n)}$. Set

$$T^{(n+1)} = T^{(n)} \cup P_{ub}, \quad B^{(n+1)} = B^{(n)} \cup \{b\}.$$

Lemma 3.13 contracts P_{ub} to a single edge of weight $\tau > 0$, and Lemma 3.14 yields

$$\widehat{S}(\tau) = S_0 + \kappa(\tau) M(v)$$

as in (3.6). The new positive eigenvalues are $\hat{\lambda}_1 < \dots < \hat{\lambda}_n$.

For $i \geq 2$, Theorem 3.8 supplies an analytic branch

$$x(\tau) = \begin{pmatrix} x_P(\tau) \\ x_b(\tau) \end{pmatrix}, \quad x_P(0) = x_P, \quad x_b(0) = 0,$$

where x_P is a nonconstant eigenfunction of S_P . By the inductive hypothesis,

$$d := \min_{k \neq l \in B^{(n)}} |x_P(k) - x_P(l)| > 0, \quad m := \min_{k \in B^{(n)}} |x_P(k)| > 0.$$

Continuity of $x_P(\tau)$ yields $\delta_1 > 0$ such that $0 \leq \tau < \delta_1$ implies

$$|x_P(\tau)_k - x_P(\tau)_l| > \frac{d}{2}, \quad |x_P(\tau)_k| > \frac{m}{2}.$$

As in Lemma 3.15, $x_b(\tau) = 0$ would force the old eigenfunction $x_P(\tau)$ to vanish at u , contradicting the inductive hypothesis for small τ . Thus $x_b(\tau) \neq 0$. Since $x_b(\tau) \rightarrow 0$, there exists $\delta_2 > 0$ such that $0 < \tau < \delta_2$ implies $|x_b(\tau)| < m/2$, and therefore

$$|x_P(\tau)_k - x_b(\tau)| \geq |x_P(\tau)_k| - |x_b(\tau)| > 0.$$

Hence $x(\tau)$ is injective on $B^{(n+1)}$ whenever $0 < \tau < \min(\delta_1, \delta_2)$.

For $\hat{\lambda}_1$, the eigenvector at $\kappa = 0$ is $c(-\mathbf{1}_n^\top, n)^\top$. Lemma 4.4 gives

$$y = -c\mathbf{1} + \kappa c(n+1) S_P^+ v + O(\kappa^2), \quad z = nc + O(\kappa^2).$$

By Lemma 4.3, the set of positive weights on $T^{(n)}$ for which $\phi := S_P^+ v$ is injective on $B^{(n)}$ has full measure. This set meets the nonempty open set on which Theorem 2.1 holds and old eigenfunctions remain injective. After such a perturbation of $w^{(n)}$,

$$d' := \min_{k \neq l \in B^{(n)}} |\phi(k) - \phi(l)| > 0,$$

and there exist $C < \infty$ and $\kappa_1 > 0$ such that $0 < \kappa < \kappa_1$ implies

$$|y_k - y_l| \geq \kappa c(n+1) d' - C\kappa^2 = \kappa(c(n+1) d' - C\kappa).$$

If moreover $C\kappa < \frac{1}{2}c(n+1) d'$, then $y_k \neq y_l$. Likewise

$$z - y_k = (n+1)c + O(\kappa),$$

so $z \neq y_k$ for all sufficiently small κ . Thus $(y^\top, z)$ is injective on $B^{(n+1)}$.

Since $\kappa = \tau$ or $\kappa = \tau/(1+\tau\alpha)$, one may choose $\tau > 0$ small enough that both the branches $i \geq 2$ and the branch $\hat{\lambda}_1$ are injective on $B^{(n+1)}$ and nowhere zero on $V(T^{(n)})$. Restoring P_{ub} by (3.4) and avoiding finitely many exceptional ratios θ_j keeps the extension nowhere zero on P_{ub} . Thus $\mathcal{P}(n+1)$ holds. At $n = N$ one obtains w_T with $P_{\text{inj}}^T(w_T) \neq 0$. $\square$

Remark 4.5. The condition excluded by Theorem 2.2(1) is $f(i) = f(j)$, equivalently $\alpha^\top f = 0$ for $\alpha = e_i - e_j$. The same compression applies to any α not parallel to $\mathbf{1}$. Theorem 2.3 records this for α with coordinates in a finite set.

Lemma 4.6. *Let $\alpha \in \mathbb{R}^B$ be not parallel to $\mathbf{1}$, write $H = \alpha^\perp$, let the columns of U form an orthonormal basis of H , and set $B_\alpha = U^\top S U$.*

- (a) *If $\alpha^\top \mathbf{1} \neq 0$, then S admits a positive-eigenvalue eigenvector in H if and only if S and B_α share a positive eigenvalue. In this case 0 is not an eigenvalue of B_α .*
- (b) *If $\alpha^\top \mathbf{1} = 0$, then $\mathbf{1} \in H$ and 0 is always a common eigenvalue; S admits a nonconstant eigenvector in H if and only if the two matrices share a further positive eigenvalue.*

Proof. Write $|B| = n$, set $q = \alpha/\|\alpha\|$, and choose an orthogonal matrix $Q = (U, q)$ whose remaining columns span H . Then $U^\top U = I_{n-1}$ and $UU^\top + qq^\top = I_n$. For any $f \in \mathbb{R}^B$ one has $\alpha^\top f = \|\alpha\| q^\top f$, so $\alpha^\top f = 0$ if and only if $f = U\xi$ for $\xi = U^\top f$.

Write $\tilde{S} = Q^\top S Q$ in blocks as

$$\tilde{S} = \begin{pmatrix} B_\alpha & y \\ y^\top & a \end{pmatrix}, \quad y = U^\top S q, \quad a = q^\top S q.$$

If $f = U\xi$ and $B_\alpha \xi = \lambda \xi$, then $Sf = \lambda f + (y^\top \xi) q$, so $Sf = \lambda f$ forces $y^\top \xi = 0$.

First suppose $\alpha^\top \mathbf{1} \neq 0$, so $\mathbf{1} \notin H$. If $B_\alpha \xi = 0$ and $f = U\xi$, then $f^\top S f = 0$, hence $Sf = 0$ by positive semidefiniteness. Connectedness gives $\ker S = \text{span}\{\mathbf{1}\}$, so f is constant, contradicting $\mathbf{1} \notin H$. Thus 0 is not an eigenvalue of B_α . As in Lemma 4.1, S admits an eigenvector in H if and only if S and B_α share an eigenvalue. That eigenvalue cannot be 0, hence is positive.

Conversely, if the two matrices share $\lambda > 0$, Cauchy interlacing applied to the block decomposition of $\tilde{S}$ forces an equality in the interlacing chain. The equality case of Theorem 3.9 produces ξ with $B_\alpha \xi = \lambda \xi$ and $y^\top \xi = 0$. Then $f = U\xi$ satisfies $Sf = \lambda f$ and $f \in H$, and $\lambda > 0$ prevents f from being constant.

Now suppose $\alpha^\top \mathbf{1} = 0$. Then $\mathbf{1} = U\xi_1$ with $\xi_1 = U^\top \mathbf{1}$, so $B_\alpha \xi_1 = 0$ and 0 is always a common eigenvalue. If $B_\alpha \xi = 0$, the same energy argument shows that $f = U\xi$ is constant, so $\ker B_\alpha$ remains one-dimensional. A nonconstant eigenvector lies in H if and only if a further common eigenvalue $\lambda > 0$ exists; Cauchy interlacing again produces the converse. $\square$

Each condition $\alpha^\top f = 0$ therefore corresponds to a resultant. The three families used below are

doubling	$\alpha = 2e_i - e_j,$	$\alpha^\top \mathbf{1} = 1,$
opposite values	$\alpha = e_i + e_j,$	$\alpha^\top \mathbf{1} = 2,$
arithmetic progression	$\alpha = e_i - 2e_j + e_k,$	$\alpha^\top \mathbf{1} = 0.$

Doubling and opposite values fall under (a) and use $\det(B_\alpha - \lambda I)$; arithmetic progressions fall under (b) and the factor $\lambda = 0$ must be removed. All three resultants are rational in w .

Positive eigenfunctions satisfy $f \perp \mathbf{1}$. If $|B| = 2$, the unique positive eigenfunction is $f = (c, -c)$ with $c \neq 0$. Opposite values $f(i) + f(j) = 0$ is the orthogonality relation and is not an extra condition; doubling forces $c = 0$, a contradiction; two points cannot form a three-term progression.

If $|B| = 3$, write the values as x, y, z with $x + y + z = 0$. Opposite values $x = -y$ forces $z = 0$, contradicting Theorem 2.1(2). A three-term progression $2y = x + z$ together with $z = -x - y$ forces $y = 0$, again contradicting Theorem 2.1(2). Doubling need not fail. The vector $(1, 2, -3)$ is orthogonal to $\mathbf{1}$, nowhere zero, injective, and satisfies $2 \cdot 1 = 2$.

Lemma 4.7. *Let T be a tree whose leaves all lie in B and $|B| = n \geq 2$. Let $v = e_u$ with $u \in B$, or $v = p_u$ with $u \in B^c$, and set $\phi = S^+ v$. If $\beta \in \mathbb{R}^B$ is not parallel to $\mathbf{1}$, then for generic positive weights $\beta^\top \phi \neq 0$.*

Proof. The cut-edge identities are (4.4)–(4.5). Fix an edge e splitting T into $T_1 \sqcup T_2$, and write $m = |B \cap T_1| \in \{1, \dots, n-1\}$, $\sigma_1 = \sum_{B \cap T_1} \beta_i$. As $w(e) \rightarrow 0^+$ with the remaining weights fixed,

$$\phi(k) - \phi(l) = \frac{c_e}{w(e)} + O(1), \quad k \in T_1 \cap B, \quad l \in T_2 \cap B,$$

where c_e is a nonzero constant if $v = e_u$. If $v = p_u$, Lemma 4.3(b) places u on one side of the cut: that side contributes the limit 0 or 1 according as u lies in T_2 or T_1 , and $c_e \rightarrow -m/n$ or $1 - m/n$, both nonzero. Since $\phi \perp \mathbf{1}$,

$$\beta^\top \phi = \frac{c_e}{w(e)} \cdot \frac{1}{n} (n\sigma_1 - m \mathbf{1}^\top \beta) + O(1).$$

The coefficient vanishes if and only if the mean of β on $T_1 \cap B$ equals $n^{-1} \mathbf{1}^\top \beta$.

If this holds for every cut, then β is constant. Otherwise let $B_* = \{b \in B : \beta(b) \neq n^{-1} \mathbf{1}^\top \beta\}$, nonempty. On the subtree spanned by B_* take a leaf $b_* \in B_*$, and let e be the edge from b_* toward the rest of B_* , with T_1 the side containing b_* . Then $T_1 \cap B$ consists of b_* together with vertices at which β already equals the global mean, so the mean on $T_1 \cap B$ equals the global mean if and only if $\beta(b_*)$ does, a contradiction. For this cut the leading term diverges, hence $\beta^\top \phi \neq 0$. $\square$

Proof of Theorem 2.3. Each condition is $\alpha^\top f = 0$ with $\alpha \in C^B$ not parallel to $\mathbf{1}$. Write $\tilde{p}_S(\lambda, w) = \det(S(w) - \lambda I)/\lambda$. If $\alpha^\top \mathbf{1} \neq 0$, Lemma 4.6(a) gives that 0 is not an eigenvalue of B_α ; set $\tilde{p}_\alpha(\lambda, w) = \det(B_\alpha(w) - \lambda I)$. If $\alpha^\top \mathbf{1} = 0$, Lemma 4.6(b) gives that 0 is always a common eigenvalue; set $\tilde{p}_\alpha(\lambda, w) = \det(B_\alpha(w) - \lambda I)/\lambda$. The finite product

$$\text{Discr}_C(w) = \prod_{\alpha} \text{Res}(\tilde{p}_S(\cdot, w), \tilde{p}_\alpha(\cdot, w)) = \frac{P(w)}{(\det L_{[B^c]})^N}$$

vanishes if and only if some positive eigenfunction meets some such α . It is enough that $P \neq 0$, and enough to produce a tree with all leaves in B at which $P \neq 0$.

Let $\mathcal{P}_C(n)$ assert the existence of a subtree with n boundary vertices and a positive weight for which Theorems 2.1 and 2.2(1) hold and every positive eigenfunction avoids every $\alpha \in C^{B(n)}$ not parallel to $\mathbf{1}$. For $n = 2$ one has $f = (c, -c)$ with $c \neq 0$, so $\alpha^\top f = c(\alpha_1 - \alpha_2) = 0$ if and only if $\alpha \parallel \mathbf{1}$.

Assume $\mathcal{P}_C(n)$, and attach a leaf b of equivalent weight $\tau > 0$ as in (3.6). Write $x = (y^\top, z)^\top$ and $\alpha^\top x = \beta^\top y + \gamma z$ with $\beta \in C^n$, $\gamma \in C$.

On a higher branch, $y(\tau) \rightarrow f_{\text{old}}$ and $z(\tau) \rightarrow 0$ with $z(\tau) \neq 0$. Thus $\alpha^\top x \rightarrow \beta^\top f_{\text{old}}$. If β is not parallel to $\mathbf{1}_n$, this is the inductive condition. If $\beta = t \mathbf{1}_n$, orthogonality gives $\alpha^\top x = (\gamma - t)z$. Since α is not parallel to $\mathbf{1}_{n+1}$, one has $\gamma \neq t$, and $z \neq 0$.

On the branch $\hat{\lambda}_1$, Lemma 4.4 gives

$$\alpha^\top x = c(n\gamma - \mathbf{1}^\top \beta) + \kappa c(n+1) \beta^\top \phi + O(\kappa^2),$$

with $\phi = S_P^+ v$. If $n\gamma \neq \mathbf{1}^\top \beta$, the constant term is nonzero. If equality holds, then β is not parallel to $\mathbf{1}$ (else $\alpha \parallel \mathbf{1}$), and Lemma 4.7 gives $\beta^\top \phi \neq 0$ after a perturbation of the old weights. For small κ , $\alpha^\top x \neq 0$.

Choosing τ small enough yields $\mathcal{P}_C(n+1)$. $\square$

4.2 A density lemma

The image of $D_w \Lambda$ is the content of Theorem 4.14. The following lemma records it for a single eigenfunction.

Lemma 4.8. *Let u be the harmonic extension of $f \in \mathbb{R}^B$, and let ψ be as in (3.3). Then*

$$\text{Im}(h \mapsto (D_w \Lambda)(h)f) = \text{span}\{(u(a) - u(b))(\psi(a) - \psi(b)) : \{a, b\} \in E\} \subseteq \mathbf{1}^\perp.$$

If G is connected and there exists a spanning tree T along whose every edge u is nonconstant, the image equals $\mathbf{1}^\perp$.

Proof. For every $w > 0$ one has $S\mathbf{1} = 0$. Differentiating along $w + \varepsilon h$ yields

$$\mathbf{1}^\top ((D_w \Lambda)(h)f) = 0.$$

Lemma 3.6 identifies the image with the displayed span.

Let T be a spanning tree with $u(a) \neq u(b)$ for all $\{a, b\} \in E(T)$. Then

$$\{\psi(a) - \psi(b) : \{a, b\} \in E(T)\} \subset \text{Im}(h \mapsto (D_w \Lambda)(h)f).$$

For $x \neq y \in B$ one has $\psi(x) = e_x$ and $\psi(y) = e_y$. Along the unique T -path $x = v_0, \dots, v_r = y$,

$$e_x - e_y = \sum_{i=0}^{r-1} (\psi(v_i) - \psi(v_{i+1})).$$

If $g \in \mathbf{1}^\perp$, then

$$\sum_{x \in B} g(x) \sum_{y \in B} (e_x - e_y) = ng,$$

so $\{e_x - e_y : x, y \in B\}$ spans $\mathbf{1}^\perp$. □

Corollary 4.9. *If $G[B]$ is connected and f is injective on B , the spanning tree in Lemma 4.8 may be taken in $G[B]$. In that case it is sufficient to vary weights of edges in $G[B]$. In particular, once Theorem 2.2(1) holds, the identity remains valid for generic weights whenever $G[B]$ is connected.*

Proof. Let T_B be a spanning tree of $G[B]$. Then $E(T_B) \subset E(G[B])$, $\psi(x) = e_x$ for $x \in B$, and $u|_B = f$. Injectivity of f gives

$$f(a) \neq f(b) \quad \text{for all } \{a, b\} \in E(T_B).$$

Lemma 4.8 applied to T_B yields the claim. □

Definition 4.10. An edge $\{a, b\} \in E$ is *flat* for $u \in \mathbb{R}^V$ if $u(a) = u(b)$.

The image of the first variation is spanned by adjacent differences. Edges outside G^B are flat for every $w > 0$.

Lemma 4.11. *Let T be a tree whose boundary contains every leaf, and let h be the harmonic extension of a Steklov eigenfunction f belonging to the eigenvalue λ . If $e = \{\alpha, \beta\}$ splits T into $T_1 \sqcup T_2$ with $\alpha \in T_1$, then*

$$w(e)(h(\alpha) - h(\beta)) = \lambda \sum_{x \in T_1 \cap B} f(x).$$

Proof. The Steklov equation is $(Lh)|_B = \lambda f$ and $(Lh)|_{B^c} = 0$. Summing over T_1 ,

$$\sum_{x \in T_1} (Lh)(x) = w(e)(h(\alpha) - h(\beta)) = \lambda \sum_{x \in T_1 \cap B} f(x).$$

□

Theorem 4.12. *Let T be a tree whose boundary B contains every leaf and $|B| \geq 2$. There exist positive edge weights for which T satisfies Theorems 2.1 and 2.2(1), and for which the harmonic extension of every nonconstant Steklov eigenfunction is nonconstant along every edge of T .*

Proof. Let $\mathcal{P}^b(n)$ assert the existence of a subtree $T^{(n)}$ with $|B^{(n)}| = n$ and a positive weight such that Theorems 2.1, 2.2(1) and 3.16 hold and

$$H(a) \neq H(b) \quad \text{for every } \{a, b\} \in E(T^{(n)})$$

and every nonconstant harmonic extension H .

If $n = 2$, Lemma 3.12 gives (3.4) with $H(x_0) = -H(x_m) = c \neq 0$, so H is strictly monotone on the path.

Assume $\mathcal{P}^b(n)$. Attach $b \in B \setminus B^{(n)}$ along P_{ub} , contracted to (3.6). On P_{ub} ,

$$H(x_j) = (1 - \theta_j)H(u) + \theta_j H(b),$$

so $\{x_{j-1}, x_j\}$ is non-flat as soon as $H(u) \neq H(b)$.

For $i \geq 2$, Theorem 3.8 supplies $H(\tau)|_{V(T^{(n)})} \rightarrow H_{\text{old}}$ with $H_{\text{old}}(a) \neq H_{\text{old}}(b)$ on every old edge and $H_{\text{old}} \neq 0$. Continuity preserves both for small τ . Moreover

$$H(\tau)(b) = x_b(\tau) \rightarrow 0, \quad H(\tau)(u) \rightarrow H_{\text{old}}(u) \neq 0,$$

so the pendant edge is non-flat.

For $\hat{\lambda}_1$, Lemma 4.11 on the contracted tree gives

$$e \text{ is flat} \iff \sigma_e(f) = \sum_{x \in T_1 \cap B^{(n+1)}} f(x) = 0.$$

The limit $f_0 = c(-\mathbf{1}_n^\top, n)^\top$ satisfies

$$\sigma_e(f_0) = \begin{cases} -cm, & b \notin T_1, \quad m = |T_1 \cap B^{(n)}| \geq 1, \\ c(n - m), & b \in T_1, \quad m \leq n - 1. \end{cases}$$

In either case $\sigma_e(f_0) \neq 0$. Continuity of σ_e excludes flat edges for small τ . Restoring P_{ub} by (3.4) completes the induction. $\square$

Lemma 4.13. *Let G be connected, $|B| \geq 2$, and $w > 0$.*

- (i) *$S(G, B, w)$ depends only on $w|_{E(G^B)}$. The harmonic extension on G^B coincides with the restriction of the extension on G .*
- (ii) *On each connected component of $G - V(G^B)$, the extension is constant, equal to its value at the unique vertex of attachment.*
- (iii) *G^B is connected. Every vertex of G^B lies on a path with both endpoints in B . Every interior vertex of G^B has degree at least two in G^B .*
- (iv) *If G admits a spanning tree whose leaves all lie in B , then $G^B = G$.*

Proof. (ii) Let K be a connected component of $G - V(G^B)$, and let N be the set of neighbours of K in $V(G^B)$. If $|N| \geq 2$, a path in K between two points of N , continued along B - B paths through those points, would place a vertex of K on a simple path with both endpoints in B . Thus $|N| = 1$. The Dirichlet problem on K has a single boundary value, so the maximum principle forces a constant. (i) The edges of K appear in $L_{[B^c]}$, but the two terms at the attachment in the harmonic equation cancel once the block is constant. Hence S and the restriction of h_f to G^B are independent of $w|_{E \setminus E(G^B)}$. (iii) A simple path in G between two vertices of B has both endpoints in B , so lies in G^B . Thus G^B is nonempty and connected. An interior vertex of degree one in G^B cannot lie on a simple B - B path: the unique incident edge cannot be traversed twice. (iv) That spanning tree is a union of B - B paths, hence $V \subset V(G^B)$. $\square$

Theorem 4.14. *Restricting to G^B , for generic edge weights the harmonic extension u of every nonconstant Steklov eigenfunction is nonconstant along every edge, and*

$$\text{Im}(h \mapsto (D_w \Lambda)(h)f) = \mathbf{1}^\perp.$$

Proof. Assume $|B| = n \geq 2$ and replace G by G^B . Let $e = \{a, b\}$ be an edge. By Lemma 4.13(iii), e lies on a path with both endpoints in B . Extending by paths to the remaining vertices of B produces a tree T_e with $e \in E(T_e)$, $V(T_e) \supset B$, and every leaf in B . Theorem 4.12 gives a weight on T_e with $H(a) \neq H(b)$ for every nonconstant extension. Let T_0 be a spanning tree of G containing T_e , and set

$$w^{(\varepsilon)}(e') = \begin{cases} w_{T_e}(e'), & e' \in E(T_e), \\ 1, & e' \in E(T_0) \setminus E(T_e), \\ \varepsilon, & e' \notin E(T_0). \end{cases} \quad (4.8)$$

At $\varepsilon = 0$ the extra edges of $T_0 \setminus T_e$ hang off T_e and do not meet B , so $S(T_0) = S(T_e)$. Thus $S(w^{(\varepsilon)}) \rightarrow S(T_e, w_{T_e})$ and $\det L_{[B^c]}(w^{(0)}) > 0$. For small $\varepsilon > 0$, one has $u(a) \neq u(b)$. Hence $\delta_e(w) = u(a) - u(b) \neq 0$.

Finitely many edges: for generic w , every edge of G^B is non-flat. A spanning tree of G^B then satisfies the hypothesis of Lemma 4.8. Combined with Lemma 3.7, each positive eigenvalue is then strictly increasing in every edge weight of G^B . $\square$

4.3 Injectivity on the vertex set

On a tree with leaves in B ,

$$H(x) - H(y) = \lambda \sum_{e \in \text{path}(x, y)} \pm \frac{\sigma_e(f)}{w(e)}, \quad \sigma_e(f) = \sum_{z \in T_1 \cap B} f(z).$$

The weights of the preceding subsection give $\sigma_e(f) \neq 0$ on every edge, so H is nonconstant along $E(T)$. Injectivity on V is the same sum being nonzero for every pair $x \neq y$. The pendant attachment (3.6) is unchanged. Along $\hat{\lambda}_1$,

$$f_0 = c(-\mathbf{1}_n^\top, n)^\top$$

is constant on the old vertices, so $\sigma_e(f_0) \neq 0$ while old values coincide. The expansion

$$H = -c\mathbf{1} + \kappa c(n+1)F + O(\kappa^2)$$

separates those values once F is injective. If the attachment vertex u lies in B , then F is the harmonic extension of S^+e_u ; if $u \in B^c$, then F is obtained by adding $L_{[B^c]}^{-1}e_u$ to the extension of S^+p_u .

The tree induction of Lemma 4.17 supplies injectivity on any tree with leaves in B . Edges outside G^B remain flat for every $w > 0$. The argument on G^B itself is the proof of Theorem 2.2(2) below.

Lemma 4.15. *Let T be a tree whose boundary contains every leaf and $|B| = n \geq 2$.*

- (a) *If $v = e_u$ with $u \in B$, then for generic positive weights the harmonic extension of $\phi = S^+v$ is injective on V .*
- (b) *If $u \in B^c$, then for generic positive weights the harmonic extension of S^+p_u is injective on V .*

Proof. Let H be the harmonic extension of $\phi = S^+v$. As in Lemma 4.3, an edge $e = \{\alpha, \beta\}$ splitting T into $T_1 \sqcup T_2$ satisfies

$$w(e)(H(\alpha) - H(\beta)) = c_e = \sum_{x \in T_1 \cap B} (v_x - \bar{v}),$$

with $m = |B \cap T_1| \in \{1, \dots, n-1\}$. For $x \neq y$,

$$H(x) - H(y) = \sum_{e' \in \text{path}(x,y)} \pm \frac{c_{e'}}{w(e')}.$$

The difference is a rational function of the edge weights.

- (a) If $v = e_u$, then $c_e = 1_{\{u \in T_1\}} - m/n$ is a nonzero constant independent of the weights. Fixing all but one weight on $\text{path}(x, y)$ leaves a non identically vanishing rational function of the remaining weight.
- (b) If $v = p_u$, take an edge e of $\text{path}(x, y)$ with T_1 the side not containing u . Then $c_e = \sigma - m/n$, where σ is the value at u of the harmonic extension of $1_{B \cap T_1}$. As $w(e) \rightarrow 0^+$ one has $\sigma \rightarrow 0$, hence $c_e \rightarrow -m/n \neq 0$. The remaining terms stay uniformly bounded, so

$$H(x) - H(y) = \frac{c_e}{w(e)} + O(1)$$

diverges and cannot vanish identically.

There are finitely many pairs $x \neq y$, so the exceptional set has measure zero. $\square$

Lemma 4.16. *Let T be a tree whose boundary contains every leaf and $|B| = n \geq 2$. Let $u \in B^c$ and $\phi = S^+p_u$. Define $K \in \mathbb{R}^V$ by $K|_B = \phi$ and, on B^c , by adding $L_{[B^c]}^{-1}e_u$ to the harmonic extension of ϕ . For generic positive weights, K is injective on V .*

Proof. Write H for the harmonic extension of ϕ and $A = L_{[B^c]}$, and set $K = H + v$ with $v|_B = 0$ and $v|_{B^c} = A^{-1}e_u$. Then $(LH)|_B = p_u - n^{-1}\mathbf{1}$ and $(LH)|_{B^c} = 0$, while $(Lv)|_{B^c} = e_u$ and $(Lv)|_B = -p_u$. Hence

$$(LK)|_B = -n^{-1}\mathbf{1}, \quad (LK)|_{B^c} = e_u.$$

Summing LK over one side T_1 of an edge $e = \{\alpha, \beta\}$ yields

$$w(e)(K(\alpha) - K(\beta)) = -\frac{m}{n} + 1_{\{u \in T_1\}},$$

with $m = |T_1 \cap B| \in \{1, \dots, n-1\}$. The right-hand side is a nonzero constant. For $x \neq y$,

$$K(x) - K(y) = \sum_{e' \in \text{path}(x,y)} \pm \frac{c_{e'}}{w(e')}.$$

Sending one weight to 0^+ makes a single term diverge while the others remain bounded, so $K(x) - K(y) \neq 0$. $\square$

Lemma 4.17. *Let T be a tree whose boundary B contains every leaf and $|B| \geq 2$. There exist positive edge weights for which T satisfies Theorems 2.1 and 2.2(1) and the harmonic extension of every nonconstant Steklov eigenfunction is injective on V .*

Proof. Let $\mathcal{P}^{\text{inj}}(n)$ assert the existence of a subtree $T^{(n)}$ with $V(T^{(n)}) \cap B = B^{(n)}$, $|B^{(n)}| = n$, and a positive weight such that Theorems 2.1, 2.2(1) and 3.16 hold and every nonconstant harmonic extension is injective on $V(T^{(n)})$.

If $n = 2$, let $T^{(2)}$ be a path in T with endpoints in B and with no intermediate boundary vertex. Lemma 3.12 gives (3.4) with $h(x_0) = -h(x_m) = c \neq 0$, so h is strictly monotone, hence injective, on the path.

Assume $\mathcal{P}^{\text{inj}}(n)$. Choose $b \in B \setminus B^{(n)}$ so that the unique path P_{ub} from b to $T^{(n)}$ meets B only at its endpoints (take an arbitrary $b_0 \in B \setminus B^{(n)}$ and retain the last vertex of $\text{path}(b_0, T^{(n)})$ still in $B \setminus B^{(n)}$). Set $T^{(n+1)} = T^{(n)} \cup P_{ub}$. Lemma 3.13 yields (3.6).

For $i \geq 2$, Theorem 3.8 supplies a branch $x(\tau)$ with $x_P(0)$ equal to an old eigenfunction whose harmonic extension H_{old} is injective and nowhere zero on $V(T^{(n)})$. Writing $H(\tau)$ for the extension of $x(\tau)$, one has $H(0)|_{V(T^{(n)})} = H_{\text{old}}$, so

$$\min_{x \neq y \in V(T^{(n)})} |H_{\text{old}}(x) - H_{\text{old}}(y)| > 0.$$

Continuity therefore yields $\tau_0 > 0$ such that $0 < \tau < \tau_0$ implies that $H(\tau)$ remains injective on $V(T^{(n)})$. Moreover $H(\tau)(b) = x_b(\tau) \rightarrow 0$ while $H(\tau)(u) \rightarrow H_{\text{old}}(u) \neq 0$, so $H(\tau)(b)$ is distinct from every old vertex for all sufficiently small τ .

For $\hat{\lambda}_1$, first perturb $w^{(n)}$ inside $\{P_s \neq 0, P_0 \neq 0, P_{\text{inj}} \neq 0\}$ so that H_ϕ is injective on $V(T^{(n)})$ if $u \in B^{(n)}$ (Lemma 4.15(a)), or so that K is injective if $u \in (B^{(n)})^c$ (Lemma 4.16). Lemma 4.4 gives

$$y = -c\mathbf{1} + \kappa c(n+1)\phi + O(\kappa^2), \quad z = nc + O(\kappa^2), \quad \phi = S_P^+ v.$$

If $u \in B^{(n)}$, the interior block is independent of τ and

$$H|_{V(T^{(n)})} = -c\mathbf{1} + \kappa c(n+1)H_\phi + O(\kappa^2).$$

If $u \in (B^{(n)})^c$, the interior block is $A + \tau ee^\top$ with $e = e_u$ and $\tau = \kappa + O(\kappa^2)$. The Sherman–Morrison formula together with $L\mathbf{1} = 0$ yields

$$H|_{V(T^{(n)})} = -c\mathbf{1} + \kappa c(n+1)K + O(\kappa^2).$$

In either case $H = -c\mathbf{1} + \kappa c(n+1)F + O(\kappa^2)$ on $V(T^{(n)})$, with F injective. Thus for $x \neq y$ in $V(T^{(n)})$,

$$H(x) - H(y) = \kappa c(n+1)(F(x) - F(y)) + O(\kappa^2),$$

while

$$z - H(x) = (n+1)c + O(\kappa).$$

Both differences are nonzero for small κ .

On the restored path $P_{ub} : x_0 = u, \dots, x_k = b$ the interpolation (3.4) applies. The identity $h(u) \neq h(b)$ makes h strictly monotone along P_{ub} . Values off the path depend on P_{ub} only through $\sum_{t=1}^k w(x_{t-1}, x_t)^{-1} = 1/\tau$. Keeping this sum fixed, the finitely many ratios

$$\theta_j = \frac{h(u)}{h(u) - h(b)}, \quad \theta_j = \frac{h(v) - h(u)}{h(b) - h(u)} \quad (v \notin P_{ub})$$

may be avoided in $(0, 1)$. The induction terminates at $n = |B|$. $\square$

Proof of Theorem 2.2(2). If $|B| = 1$, only constants occur. Nowhere vanishing is Theorem 3.16. By Lemma 4.13 it is enough to prove injectivity after replacing G by G^B , so henceforth $G = G^B$ and $|B| \geq 2$.

Fix $x \neq y \in V$. If $x \in B$ then $h(x) = f(x)$; if $x \in B^c$ then (1.10) gives $h(x) = p_x^\top f$. Thus $h(x) - h(y) = \alpha^\top f$, where

$$\alpha = \begin{cases} e_x - e_y, & x, y \in B, \\ p_x - e_y, & x \in B^c, y \in B, \\ p_x - p_y, & x, y \in B^c. \end{cases}$$

As in (1.7) and (1.10), α is rational in the edge weights, with denominator a power of $\det L_{[B^c]}$. The three cases satisfy $\alpha \perp \mathbf{1}$. Write

$$\alpha(w) = \frac{N_{xy}(w)}{(\det L_{[B^c]})^k}$$

with N_{xy} a polynomial vector. On $\{N_{xy} \neq 0\}$, the vector α is not parallel to $\mathbf{1}$. Replacing $e_i - e_j$ by α in Lemma 4.2, a nonconstant eigenfunction satisfies $\alpha^\top f = 0$ if and only if S and the compression B_α share a positive eigenvalue. The corresponding resultant is $Q_{xy}(w)/(\det L_{[B^c]})^N$. If there is a nonempty open set of weights on which every nonconstant extension satisfies $u(x) \neq u(y)$, then $N_{xy} \neq 0$ and $Q_{xy} \neq 0$.

Suppose first that some path P with both endpoints in B meets both x and y . Attach each remaining $b \in B$ along a path meeting the current tree at a single vertex. The resulting tree T_* satisfies $V(T_*) \supset B \cup \{x, y\}$ and has all leaves in B . Lemma 4.17 supplies w_* on T_* such that every nonconstant harmonic extension H is injective on $V(T_*)$, hence $H(x) \neq H(y)$. Let T_0 be a spanning tree of G containing T_* , and set

$$w^{(\varepsilon)}(e) = \begin{cases} w_*(e), & e \in E(T_*), \\ 1, & e \in E(T_0) \setminus E(T_*), \\ \varepsilon, & e \notin E(T_0). \end{cases}$$

At $\varepsilon = 0$ the extra edges of $T_0 \setminus T_*$ hang off T_* and do not meet B , so $S(T_0) = S(T_*)$. Thus $S(w^{(\varepsilon)}) \rightarrow S(T_*, w_*)$ and $\det L_{[B^c]}(w^{(0)}) > 0$, whence $u(x) \neq u(y)$ for small $\varepsilon > 0$.

If no such path exists, then $x, y \in B^c$. Indeed, if $y \in B$ and P is a B - B path through x , concatenating a path from y to P with the arc of P through x yields a B - B path through both. Let P be a path through x with both endpoints in B . Delete $V(P)$, write K for the component containing y , and N for the vertices of P adjacent to K . Connectedness of G gives $N \neq \emptyset$.

If $|N| = 1$, say $N = \{\alpha\}$, and K meets B at some b , then a path from b through y to α , continued along P to the endpoint on the side of x , is a B - B path through x and y , a contradiction. Thus $K \cap B = \emptyset$. Every path from y to B then meets α , so y lies on no simple B - B path, contradicting $G = G^B$. Hence $|N| \geq 2$.

Take distinct $\alpha_1, \alpha_2 \in N$ and a path in K through y from α_1 to α_2 . Together with the arc of P between α_1 and α_2 this yields a cycle C . If that arc does not contain x , then

one endpoint of P — x — α_1 — K (through y) — α_2 — the other endpoint

is a B - B path through x and y , a contradiction. Thus x lies on the arc.

The same trunk construction produces a tree T_* with all leaves in B , $V(T_*) \supset B \cup \{x, \alpha_1, \alpha_2\}$, and $T_* \cap K = \emptyset$. Lemma 4.17 gives $H(\alpha_1) \neq H(\alpha_2)$. Let $Q_y = (\alpha_1 = q_0, \dots, q_r = y, \dots, q_m = \alpha_2)$

be the arc of C through y , and write $G_* = T_* \cup Q_y$. For $\tau > 0$, scale the weights on Q_y by τ with the ratios of the terms w^{-1} fixed, so that

$$\theta = \frac{\sum_{t=1}^r w(q_{t-1}, q_t)^{-1}}{\sum_{t=1}^m w(q_{t-1}, q_t)^{-1}} \in (0, 1)$$

satisfies

$$(1 - \theta)H(\alpha_1) + \theta H(\alpha_2) \neq H(x).$$

At most one θ is excluded per eigenfunction. The Schur complement on the interior of Q_y replaces Q_y by an edge of weight $O(\tau)$ between α_1 and α_2 , and

$$u(y) = (1 - \theta)u(\alpha_1) + \theta u(\alpha_2).$$

As $\tau \rightarrow 0$, one has $S(G_*, \tau) \rightarrow S(T_*, w_*)$ and

$$u(y) \rightarrow (1 - \theta)H(\alpha_1) + \theta H(\alpha_2) \neq H(x).$$

Assigning weight ε to $E \setminus E(G_*)$ and sending $\varepsilon \rightarrow 0$ preserves $u(x) \neq u(y)$ for small ε .

Thus $N_{xy} \not\equiv 0$ and $Q_{xy} \not\equiv 0$. The finite union, over pairs $x \neq y$, of the sets $\{N_{xy} = 0\} \cup \{Q_{xy} = 0\}$ has measure zero. $\square$

Remark 4.18. If $G \neq G^B$, then h is constant on each hanging block for every $w > 0$, and injectivity on V fails. Theorem 2.2(2) gives injectivity on $V(G^B)$. If G admits a spanning tree with all leaves in B , then $G^B = G$. The converse fails on C_4 with B a pair of opposite vertices, Figure 3: here $G^B = G$, yet every spanning tree has a leaf outside B . A nonconstant eigenfunction is a multiple of $(1, -1)$ on B , and the two interior vertices are convex combinations of the endpoints; they are distinct for generic weights. If $B = V$, then $G^B = G$, and the statement includes the graph Laplacian.

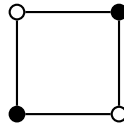


Figure 3: Open vertices lie in B , filled vertices in B^c . Every simple B – B path uses both interior vertices, so $G^B = G$. Every spanning tree has a leaf in B^c .

Equal weights may lie in the exceptional set. On a Y-shaped tree with three equal arms, λ_1 has multiplicity two and an eigenfunction may agree on two leaves. On C_4 with $B = V$ and equal weights, the first positive eigenspace contains vectors that agree on opposite vertices. Both lie in $\{P_s = 0\}$ or in some $\{Q_{xy} = 0\}$. On a path, injectivity on V holds for all $w > 0$.

4.4 Vertex Morse and Poincaré–Hopf

Closed stars in G^B lie in $V(G^B)$. Theorem 2.2(2) therefore yields vertex Morse on G^B .

Lemma 4.19 (Theorem 1, [8]). *Let $G = (V, E)$ be a finite simple graph and f a vertex Morse function. Then*

$$\sum_{v \in V} i_f(v) = \chi(G).$$

Proof. Order the vertices $v_1, \dots, v_m$ by increasing values of f . Closed-star injectivity forbids adjacent equal values, so the order among equal values is immaterial. Let G_j be the induced subgraph on $\{v_1, \dots, v_j\}$, and $G_0 = \emptyset$. It is enough that

$$\chi(G_j) - \chi(G_{j-1}) = i_f(v_j).$$

The neighbour set of v_j in G_j is $\text{lk}^-(v_j)$. New cliques are $\{v_j\}$ and $\{v_j\} \cup \sigma$ for cliques $\sigma \subset \text{lk}^-(v_j)$, hence

$$c_1(G_j) = c_1(G_{j-1}) + 1, \quad c_{k+1}(G_j) = c_{k+1}(G_{j-1}) + c_k(\text{lk}^-(v_j)) \quad (k \geq 1),$$

and

$$\chi(G_j) - \chi(G_{j-1}) = 1 + \sum_{k \geq 1} (-1)^k c_k(\text{lk}^-(v_j)) = 1 - \chi(\text{lk}^-(v_j)) = i_f(v_j).$$

□

Proof of Theorem 2.4. Theorem 2.2(2) gives that u is injective on $V(G^B)$. Every closed star of G^B is contained in $V(G^B)$, so u is vertex Morse on G^B . Lemma 4.19 yields

$$\sum_{v \in V(G^B)} i_u(v) = \chi(G^B).$$

□

Forman Morse fails generically, as recorded in Subsection 1.2, and is not used.

4.5 Nodal domains on the support graph

Lemma 4.20. *Let $w > 0$ and $|B| = n$, and let G_S be as in Definition 1.9.*

- (i) *As a combinatorial graph, G_S is independent of the choice of positive weights;*
- (ii) *if G is connected then G_S is connected;*
- (iii) *$S\mathbf{1} = 0$, so $S_{xx} = -\sum_{y \neq x} S_{xy}$. Thus S is the Laplacian of G_S with edge weights $-S_{xy}$. The diagonal is nonnegative, the off-diagonal is nonpositive, and $\ker S = \text{span}\{\mathbf{1}\}$. If $n \geq 2$ the diagonal is strictly positive;*
- (iv) *Steklov eigenpairs are eigenpairs of this weighted Laplacian. In particular, if $B = V$ then $G_S = G$ and $S = L$.*

Proof. For $x \neq y \in B$, (1.9) gives

$$-S_{xy} = w(x, y) + \sum_{z, t \in B^c} w(x, z) (L_{[B^c]}^{-1})_{zt} w(t, y) \geq 0,$$

with strict inequality if and only if $\{x, y\} \in E$ or there is a path from x to y with all intermediate vertices in B^c . Both alternatives depend only on (G, B) , which is (i).

If G is connected and $x = v_0, \dots, v_m = y$ is a path in G , each pair of successive visits to B is an edge of G_S . Concatenating these edges yields a path in G_S , which is (ii).

Proposition 1.7 gives $S\mathbf{1} = 0$, hence

$$S_{xx} = \sum_{y: S_{xy} \neq 0} (-S_{xy}).$$

If $Sf = 0$, then

$$f^\top Sf = \frac{1}{2} \sum_{x,y:S_{xy} \neq 0} (-S_{xy})(f(x) - f(y))^2 = 0,$$

so f is constant on every edge of G_S . Connectedness forces f constant, and $\ker S = \text{span}\{\mathbf{1}\}$. If $n \geq 2$, every vertex of G_S is incident to an edge, so $S_{xx} > 0$. This is (iii). Assertion (iv) is $Sf = \lambda f$; if $B = V$ then $S = L$ and $G_S = G$. $\square$

Lemma 4.21. *The graph G_S is obtained from $G[B]$ by adding, for each connected component C of $G[B^c]$, a clique on the neighbour set $N(C) \subset B$. Thus $G[B]$ is a spanning subgraph of G_S , and*

$$G_S \text{ is complete} \iff \forall x \neq y \in B, \quad \{x, y\} \in E(G[B]) \text{ or } x, y \in N(C) \text{ for some } C.$$

Proof. By Lemma 4.20, $\{x, y\} \in E(G_S)$ if and only if $\{x, y\} \in E(G[B])$ or there is a path from x to y with interior in B^c . In the latter case the interior lies in a single component C of $G[B^c]$ and $x, y \in N(C)$. Conversely, if $x, y \in N(C)$, a path from x into C and on to y has interior in B^c . $\square$

Proof of Proposition 2.5. Every edge of $G[B]$ is an edge of G_S , so for every $w > 0$

$$\{x \in B : f(x) \geq f(y) \text{ for all } y \sim_{G_S} x\} \subset \{x \in B : f(x) \geq f(y) \text{ for all } y \sim_{G[B]} x\},$$

and likewise for local minima.

If $G_S = K_{|B|}$ and f is injective on B , a local maximum is a global maximum, and

$$x_+ = \arg \max_B f, \quad x_- = \arg \min_B f$$

are unique. Injectivity holds for generic w by Theorem 2.2(1). If $G[B]$ is complete, Lemma 4.21 implies that G_S is complete, and the same pair $\{x_+, x_-\}$ is the set of extrema on both graphs. $\square$

Theorem 4.22 (Theorem (3,3), [7]). *Let $H = (V_H, E_H)$ be a finite connected graph with positive edge weights, and let L be its weighted Laplacian. Then 0 is the simple least eigenvalue of L , with $\ker L = \text{span}\{\mathbf{1}\}$. If $f : V_H \rightarrow \mathbb{R}$ belongs to the second-smallest eigenvalue, the induced subgraphs*

$$H[\{x \in V_H : f(x) \geq 0\}] \quad \text{and} \quad H[\{x \in V_H : f(x) \leq 0\}]$$

are both connected.

Proposition 4.23. *For generic weights, λ_1 is simple. A corresponding eigenfunction f has the property that $\{x \in B : f(x) \geq 0\}$ and $\{x \in B : f(x) \leq 0\}$ induce connected subgraphs of G_S . Moreover*

$$(Sf)(x) = \sum_{y:S_{xy} \neq 0} (-S_{xy})(f(x) - f(y)) = \lambda_1 f(x),$$

so $f > 0$ at a local maximum and $f < 0$ at a local minimum. In particular, if G_S itself is a cycle or a path and f is injective, then f has exactly two local extrema on G_S , one maximum and one minimum.

Proof. By Lemma 4.20, S is the weighted Laplacian of the connected graph G_S with weights $-S_{xy} > 0$. Thus 0 is the least eigenvalue and λ_1 is the second-smallest. For generic weights λ_1 is simple, and Theorem 4.22 yields that $G_S[\{f \geq 0\}]$ and $G_S[\{f \leq 0\}]$ are connected.

If x is a local maximum on G_S and $f(x) \neq f(y)$ for all $y \sim_{G_S} x$ (Theorem 2.2(1)), then each summand $(-S_{xy})(f(x) - f(y))$ is positive, so $\lambda_1 f(x) > 0$ and $f(x) > 0$. Likewise $f(x) < 0$ at a local minimum.

Now suppose G_S is a cycle and f is injective. Then local maxima lie in $\{f > 0\}$ and local minima in $\{f < 0\}$, while $\{f \geq 0\}$ is a connected arc. If that arc contained two local maxima x_i, x_j , the path in the arc joining them would contain a local minimum p with $f(p) \geq 0$, contradicting $f < 0$ at a local minimum. Thus there is a unique local maximum and a unique local minimum. If G_S is a path, the endpoints are local extrema and $\{f \geq 0\}$ is a subpath; the same comparison excludes two local maxima, so the extrema are precisely the two endpoints. $\square$

Corollary 4.24. *For generic weights, λ_1 is simple and the corresponding eigenfunction is nowhere zero on B . Hence $\{f > 0\} = \{f \geq 0\}$ and $\{f < 0\} = \{f \leq 0\}$, and both sets induce connected subgraphs of G_S . Thus λ_1 has exactly two nodal domains on G_S .*

By Proposition 2.5,

$$\{\text{local extrema of } f \text{ on } G_S\} \subset \{\text{local extrema of } f \text{ on } G[B]\}.$$

The converse need not hold. If $G_S = K_{|B|}$ and f is injective, the left-hand side has cardinality two. If G_S is a tree, every leaf is a local extremum, and Theorem 2.6(3) gives equality for λ_1 . Thus λ_1 having exactly two local extrema on B is not generic for arbitrary (G, B) . It holds when G_S is a cycle, a path, or a complete graph.

Lemma 4.21:

$$G_S = G[B] \iff N(C) \text{ is a clique in } G[B] \text{ for every interior component } C$$

(including $|N(C)| \leq 1$).

Example 4.1. *A wheel with hub in B^c and rim B , as in Figure 4, has a single interior component C with $N(C) = B$, hence $G_S = K_{|B|}$. Every nonconstant eigenfunction has exactly two local extrema on G_S , while $G[B]$ is a cycle and may have additional peaks.*

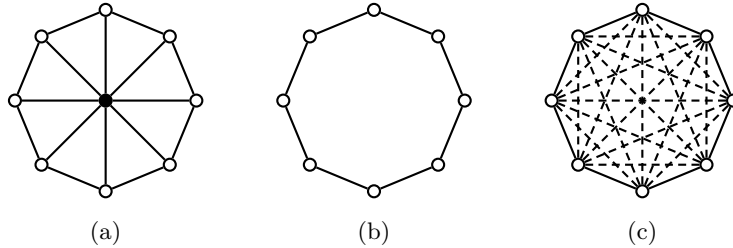


Figure 4: Open vertices lie in B , the filled hub in B^c . (a) The graph G . (b) The induced graph $G[B]$ is an 8-cycle. (c) The support graph $G_S = K_8$: dashed edges are the clique edges not already in $G[B]$.

Example 4.2. *Let $G[B] = C_8 = (x_0, \dots, x_7)$ and add interior paths*

$$x_0 - p_1 - p_2 - x_4, \quad x_2 - q_1 - q_2 - x_6,$$

as in Figure 5(a). Assign weight 5 to the edges $\{x_{2k}, x_{2k+1}\}$ and weight $1/5$ to $\{x_{2k+1}, x_{2k+2}\}$ (indices modulo 8), and weight 3 to the interior-path edges. The weights have a $\pi/2$ rotational symmetry, so S has a repeated positive eigenvalue and the example is not generic. By the symmetry one may take a first positive eigenfunction with

$$\text{sign}(f|_B) = (+, +, -, -, +, +, -, -).$$

The two positive arcs are joined by an edge of G_S , Figure 5(c), so $G_S[\{f > 0\}]$ is connected, while $G[B][\{f > 0\}]$ is not, Figure 5(b). Each interior vertex has degree two, hence takes a convex combination of its path-neighbours, and u has two nodal domains on V . Both interior components satisfy $|N(C)| = 2$, as in Theorem 2.7.

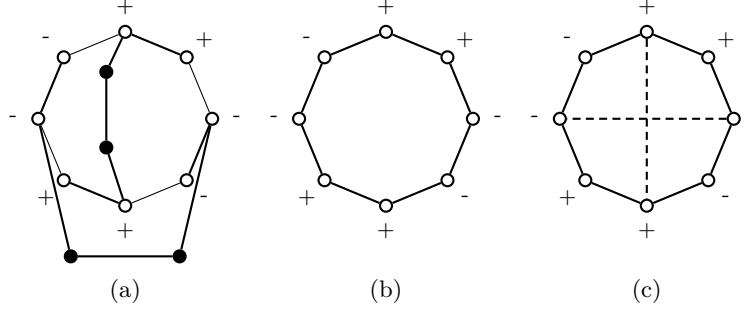


Figure 5: Open vertices lie in B , filled vertices in B^c . Signs are those of f on B . (a) The graph G : thick cycle edges have weight 5, thin edges weight $1/5$, interior-path edges weight 3. The path $x_2-q_1-q_2-x_6$ is drawn outside the cycle so that the two interior components do not cross; combinatorially both are interior paths in B^c . (b) The induced graph $G[B]$: the two positive arcs are disconnected. (c) The support graph G_S : dashed edges are the cliques on $N(C_p) = \{x_0, x_4\}$ and $N(C_q) = \{x_2, x_6\}$.

Fiedler's theorem applies only to the second-smallest eigenvalue. If $Sf = \lambda_k f$ with $k \geq 2$, neither $G_S[\{f \geq 0\}]$ nor $G_S[\{f \leq 0\}]$ need be connected. On a general graph the number of local extrema is not a function of k ; the cycle and the path are treated separately in Subsection 4.6.

4.6 Critical points on cycles and paths

Proof of the first assertion of Theorem 2.6. Theorem 2.2(1) gives injectivity on B . Closed stars of $G[B]$ and of G_S lie in B , so f is vertex Morse on both. Lemma 4.19 on G_S yields

$$\sum_{v \in B} i_f(v) = \chi(G_S).$$

□

Proposition 4.25. *Suppose $G[B]$ is a cycle and f is injective on B . Then f is vertex Morse on $G[B]$. The cycle has at least two local extrema, and maxima and minima alternate.*

Proof. Label the cycle $x_0, \dots, x_{n-1}$ with indices modulo n and $n \geq 3$. The closed star of x_i is $\{x_{i-1}, x_i, x_{i+1}\}$ (or B if $n = 3$). Injectivity on B restricts to each closed star, so f is vertex Morse.

Let x_+ (resp. x_-) be a global maximum (resp. minimum). Injectivity gives $x_+ \neq x_-$ and

$$f(x_+) > f(y) \quad \text{for all } y \sim x_+, \quad f(x_-) < f(y) \quad \text{for all } y \sim x_-.$$

Suppose two consecutive local extrema x_i, x_j are both maxima, with no extremum on the open arc between them. Adjacent vertices cannot both be maxima, so the arc is a path

$$x_i = p_0, p_1, \dots, p_\ell = x_j, \quad \ell \geq 2,$$

with $f(p_1) < f(p_0)$ and $f(p_{\ell-1}) < f(p_\ell)$. A minimiser p_s of f on this path satisfies $1 \leq s \leq \ell - 1$ and

$$f(p_s) < f(p_{s-1}), \quad f(p_s) < f(p_{s+1}),$$

hence is a local minimum, a contradiction. Replacing f by $-f$ excludes consecutive minima. $\square$

Theorem 4.26. *Suppose $G[B]$ is a cycle and $|B| \geq 3$. For generic edge weights, every nonconstant Steklov eigenfunction f is vertex Morse on $G[B]$, and*

- (i) *every local extremum is strict, maxima and minima alternate along the cycle, and the extremal values are pairwise distinct;*
- (ii) *if $|B| \geq 4$, then $2f(x_i) \neq f(x_{i-1}) + f(x_{i+1})$ at every vertex of the cycle.*

If $G[B]$ is a path, the restriction is likewise vertex Morse, and both endpoints are local extrema.

Proof. Theorem 2.2(1) and Proposition 4.25 give vertex Morse, alternating extrema, and

$$\#\{\text{local maxima}\} \geq 1, \quad \#\{\text{local minima}\} \geq 1.$$

Injectivity forces distinct extremal values, which is (i). If $|B| \geq 4$, the vertices x_{i-1}, x_i, x_{i+1} are pairwise distinct, and Theorem 2.3(iii) is (ii).

If $G[B]$ is a path, closed stars have two or three vertices, so injectivity yields vertex Morse, and each endpoint is a local extremum. $\square$

Write $\varphi(f)$ for the number of edges of G_S on which f changes sign.

Lemma 4.27. *Suppose G_S is a cycle or a path. For generic w , a nonconstant eigenfunction f is injective and nowhere zero on B . Then every local maximum satisfies $f > 0$, every local minimum satisfies $f < 0$, and each nodal domain contains exactly one local extremum. Hence the number of local extrema equals the number of nodal domains. On a cycle that number is even.*

Proof. Theorems 2.1 and 2.2(1) give injectivity and nonvanishing. For $\lambda > 0$,

$$\lambda f(x) = \sum_{y: S_{xy} \neq 0} (-S_{xy})(f(x) - f(y)).$$

At a local maximum the summands are nonnegative, and injectivity makes them positive, so $f(x) > 0$. Likewise $f(x) < 0$ at a local minimum.

A positive nodal domain is an arc (or a subpath). Two local maxima would force a local minimum p on the path between them with $f(p) \geq 0$. Thus each positive domain has at most one local maximum, and each negative domain at most one local minimum.

Let x_* maximise f on a positive domain D . Then x_* is a local maximum on G_S . If x_* is interior to D , both G_S -neighbours lie in D and take strictly smaller values. If x_* is an endpoint of D , the neighbour in D is smaller, and either x_* has degree one on the path G_S , or the unique neighbour outside D has the opposite sign. Negative domains are the same. On a cycle, $\varphi(f)$ is even. $\square$

Proof of Theorem 2.6(1). If G_S is a path, Lemma 4.20 gives that S is tridiagonal in the path order, with subdiagonal entries $-S_{i,i+1} > 0$ for every $w > 0$. A symmetric irreducible tridiagonal matrix has simple spectrum. Write $0 = \mu_1 < \mu_2 < \dots < \mu_n$ for the eigenvalues of S , so $\lambda_j = \mu_{j+1}$. The discrete Sturm theorem for a Jacobi matrix with negative subdiagonal—the Laplacian or Steklov matrix of a path—asserts that an eigenfunction belonging to μ_k has exactly $k - 1$ sign-changing edges. (Oscillatory Jacobi matrices in Gantmacher–Krein [13] have the opposite, positive,

subdiagonal convention.) Thus $\varphi(f) = j$ for an eigenfunction of λ_j . Removing j edges from a path yields $j + 1$ components, hence $j + 1$ nodal domains. Lemma 4.27 gives $j + 1$ local extrema. Both endpoints have degree one, hence are extrema. For $j = 1$ the two extrema are the endpoints, so f is strictly monotone. $\square$

Proof of Theorem 2.6(2). Number the eigenvalues of S as $0 = \mu_1 < \mu_2 \leq \dots$. Berkolaiko [11] gives $\varphi \geq k - 1$ whenever μ_k is simple and the eigenvector is nowhere zero. Colin de Verdière [12] identifies the nodal defect $\varphi - (k - 1)$ with the number of negative eigenvalues of a magnetic perturbation, whence $\varphi \leq k - 1 + \beta$ with $\beta = |E| - |V| + 1$. Thus

$$k - 1 \leq \varphi \leq k - 1 + \beta.$$

Here G_S is a cycle, so $\beta = 1$. For λ_j one has $k = j + 1$, hence $j \leq \varphi \leq j + 1$. On a cycle, φ is even and $\varphi \geq 2$ for $j \geq 1$. If j is even the unique even integer in $[j, j + 1]$ is j ; if j is odd it is $j + 1$. Both equal $2\lceil j/2 \rceil$. Lemma 4.27 gives the same number of extrema, which therefore alternate along the cycle. For $j = 1$ the count is two. $\square$

Proof of Theorem 2.6(3). Theorems 2.1 and 2.2(1) give that λ_1 is simple, and that f is nowhere zero and injective on B . Corollary 4.24 gives that $G_S[\{f > 0\}]$ and $G_S[\{f < 0\}]$ are connected. Since G_S is a tree, the two complementary connected induced subgraphs are joined by a unique edge $e^* = \{\alpha, \beta\}$. Take $f(\alpha) > 0 > f(\beta)$, and write T_+ (resp. T_-) for the component of $G_S - e^*$ containing α (resp. β).

Lemma 4.20 makes S the weighted Laplacian of G_S . For an edge $e = \{\alpha', \beta'\}$ of G_S , splitting G_S into $T_1 \sqcup T_2$ with $\alpha' \in T_1$, summing $Sf = \lambda_1 f$ over T_1 leaves only the crossing edge:

$$w_e(f(\alpha') - f(\beta')) = \lambda_1 \sum_{x \in T_1} f(x), \quad w_e := -S_{\alpha'\beta'} > 0.$$

Let $x \in T_+ \setminus \{\alpha\}$ and let p be the neighbour of x toward α . Taking T_1 to be the component of $G_S - e$ containing x , where $e = \{p, x\}$, one has $T_1 \subset T_+$, hence $f > 0$ on T_1 . The right-hand side is positive, so $f(x) > f(p)$. Along the unique path from α to any vertex of T_+ , the values therefore increase. The same argument on T_- yields that f decreases (becomes more negative) along every path from β to a vertex of T_- .

A leaf has degree one, hence is a local extremum. If v is not a leaf, then $\deg(v) \geq 2$. If $v \in T_+ \setminus \{\alpha\}$, the parent has a strictly smaller value and some child has a strictly larger value. If $v = \alpha$ and α is not a leaf, then α has a neighbour in T_+ with a larger value and is adjacent to β with $f(\beta) < 0$. Interior vertices of T_- are likewise not extrema. Thus the local extrema are exactly the leaves, Figure 6. $\square$

Lemma 4.28. *Let C be a connected component of $G[B^c]$ with $N(C) = \{x, y\}$, and let $w > 0$. Let φ be the unique function on $C \cup \{x, y\}$ that is harmonic on C with $\varphi(x) = 1$ and $\varphi(y) = 0$. Then $0 < \varphi < 1$ on C . For every $t \in (0, 1)$, the induced subgraph on $\{\varphi > t\}$ is connected and contains x , and $\{\varphi < t\}$ is connected and contains y . If $v \in C$ and $\varphi(v) \neq t$, there is a path from v to x or to y along which φ is strictly monotone and remains on the same side of t .*

Proof. The function φ is not constant. A nonconstant function harmonic on the interior of a connected graph with positive weights attains its extrema only on $\{x, y\}$. Thus $0 < \varphi < 1$ on C .

Let U be a component of $\{\varphi > t\}$, and let v_* maximise φ on U . Then $\varphi(v_*) > t$, and $\varphi \leq \varphi(v_*)$ at every neighbour of v_* . If $v_* \in C$, harmonicity forces equality at every neighbour. Neighbours

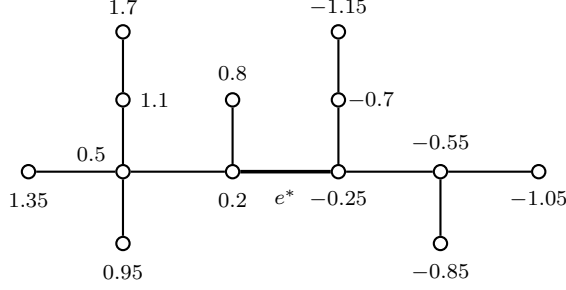


Figure 6: A tree G_S and a first positive eigenfunction. The unique sign-changing edge e^* is drawn thicker. Values increase along T_+ away from e^* and decrease along T_- . Local extrema are the leaves.

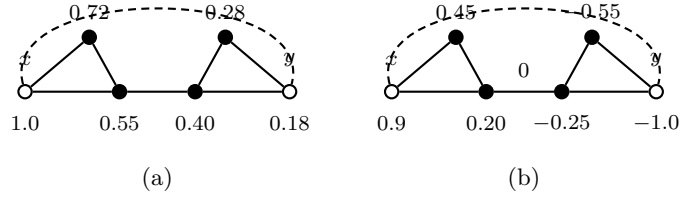


Figure 7: A two-pole. Open vertices lie in B , filled vertices in B^c ; the dashed edge belongs to G_S . Interior values are convex combinations of the poles. (a) Same sign: u keeps the sign on C . (b) Opposite signs: a unique zero of u on the interior bridge, not attained at a vertex.

outside U have $\varphi \leq t < \varphi(v_*)$, so v_* has none. The equality set fills U and U has no leaving edge, contradicting connectedness of $C \cup \{x, y\}$. Hence $v_* = x$, and $\{\varphi > t\}$ is connected. The set $\{\varphi < t\}$ is $1 - \varphi$.

If $v \in C$ and $\varphi(v) < 1$, some neighbour has a strictly larger value: otherwise the mean value forces a plateau, then constancy on C . Iterating yields a strictly increasing path to x . If $\varphi(v) > t$, the path stays in $\{\varphi > t\}$. A decreasing path to y is the same. $\square$

Proof of Theorem 2.7. Theorems 2.1 and 2.2: for generic w , f is injective and nowhere zero on B , and u vanishes at no vertex of V . Nodal domains of u are the components of $G[\{u > 0\}]$ and $G[\{u < 0\}]$; likewise for f on G_S .

Let C be an interior component. If $|N(C)| = 1$, then C lies outside G^B , Lemma 4.13, and $u|_C$ is constant. No new nodal domain. If G is connected and $B \neq \emptyset$, then $|N(C)| \neq 0$.

Now let $N(C) = \{x, y\}$. Lemma 4.21: $\{x, y\} \in E(G_S)$. Lemma 4.28: on $C \cup \{x, y\}$,

$$u = f(y) + (f(x) - f(y))\varphi, \quad 0 < \varphi < 1 \text{ on } C,$$

with $\varphi(x) = 1$, $\varphi(y) = 0$, Figure 7.

If $f(x)f(y) > 0$, then u lies strictly between $f(x)$ and $f(y)$, so u keeps the sign of the poles on C . The edge $\{x, y\}$ is not sign-changing, so x and y already lie in the same nodal domain of f on G_S .

If $f(x) > 0 > f(y)$, then $u = 0$ at the unique level

$$\varphi = t_* = \frac{f(y)}{f(y) - f(x)} \in (0, 1).$$

Generically t_* is not attained at a vertex. Lemma 4.28: $\{\varphi > t_*\}$ is connected and contains x , while $\{\varphi < t_*\}$ is connected and contains y . The level $\{\varphi = t_*\}$ is a cut of C (a single edge in Figure 7(b); in general a cut-set) and realises the already counted sign-changing edge of G_S .

Edges of $G[B]$ are edges of G_S . Hence the nodal domains of u on V are in bijection with those of f on G_S .

In particular, for λ_1 , Corollary 4.24 gives two nodal domains of f on G_S , hence two of u on V and on $V(G^B)$. $\square$

The hypothesis $|N(C)| \leq 2$ is used to realise each edge of G_S by a path meeting B only at its endpoints. If some C meets B in three or more vertices, a path in G realising an edge of G_S may pass through a third pole, and the interpolation need not preserve sign. Wheels are of this type. Stars with at least three leaves have $|N(C)| \geq 3$; the first eigenvalue is computed in Theorem 4.30.

Corollary 4.29. *Under the hypothesis of Theorem 2.7, for generic w , the local extrema of an eigenfunction belonging to λ_1 on G^B are in bijection with the local extrema of f on G_S . If moreover G_S is a tree, these extrema are exactly the leaves of G_S .*

Proof. Harmonicity forbids a strict local extremum at any interior vertex. Theorem 2.2(2) gives injectivity on $V(G^B)$, so there are no plateaus. Local extrema of u on G^B therefore lie in B .

On a component with $N(C) = \{x, y\}$,

$$u = f(y) + (f(x) - f(y))\varphi \quad \text{on } C, \quad 0 < \varphi < 1.$$

Interior values lie strictly between $f(x)$ and $f(y)$. Neighbours of x in G^B are neighbours in $G[B] \subset G_S$, or interior vertices of a two-pole. Thus x is a local maximum of u on G^B if and only if $f(x) > f(y)$ for every G_S -neighbour y . Minima are the same. Theorem 2.6(3) gives the last sentence. $\square$

Theorem 4.30. *Let G be a star whose hub lies in B^c and whose leaves are $B = \{1, \dots, n\}$, $n \geq 2$, with spoke weights $w_i > 0$. Write $W = \sum_i w_i$ and $w = (w_1, \dots, w_n)^\top$. Then*

$$S = \text{diag}(w) - \frac{ww^\top}{W},$$

and the value at the hub is $w^\top f/W$. If the spoke weights are pairwise distinct, ordered $w_{(1)} < \dots < w_{(n)}$, then $\lambda_1 \in (w_{(1)}, w_{(2)})$, exactly one leaf has sign opposite to the hub, and the harmonic extension has exactly two nodal domains on V , Figure 8.

Proof. One has $S\mathbf{1} = 0$. For $\lambda > 0$ and $Sf = \lambda f$, set $c := w^\top f/W$. The identity $w_i(f_i - c) = \lambda f_i$ rearranges to $f_i = cw_i/(w_i - \lambda)$ whenever $c \neq 0$ and $\lambda \neq w_i$. Orthogonality $\mathbf{1}^\top f = 0$ becomes

$$\sum_{i=1}^n \frac{w_i}{w_i - \lambda} = 0.$$

The left-hand side has a simple pole at each w_i , is strictly positive on $(0, w_{(1)})$, and has positive derivative $\sum_i w_i/(w_i - \lambda)^2$ on each interval between consecutive poles. The first positive root therefore lies in $(w_{(1)}, w_{(2)})$. Exactly one index then satisfies $w_i < \lambda_1$, so that leaf is opposite in sign to c , while the remaining leaves have the same sign as c . The hub equals c . Leaves of the majority sign are joined through the hub and form one nodal domain; the unique opposite leaf forms the other.

If $c = 0$ and the w_i are distinct, then $w_i f_i = \lambda f_i$ for each i . If λ equals no w_i , then $f = 0$. If $\lambda = w_j$, then $f_i = 0$ for $i \neq j$, and $\mathbf{1}^\top f = f_j = 0$, so again $f = 0$. $\square$

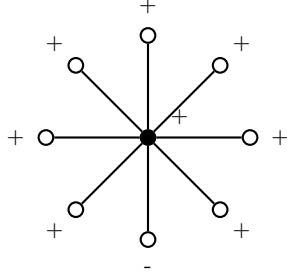


Figure 8: A star with hub in B^c . Open vertices are the leaves B . For distinct spoke weights, exactly one leaf is opposite in sign to the hub; the remaining leaves join through the hub and form one nodal domain.

Example 4.3. Let G be the cycle $v_0 \cdots v_5$ with $B = V$ and consecutive edge weights $1, 2, \frac{1}{2}, 3, \frac{2}{5}, \frac{3}{2}$. A first positive eigenfunction is, to three decimals,

$$(-0.380, -0.314, -0.183, 0.568, 0.575, -0.266),$$

as in Figure 9. The unique local maximum is v_4 and the unique local minimum is v_0 . There are two sign changes, hence two nodal domains.

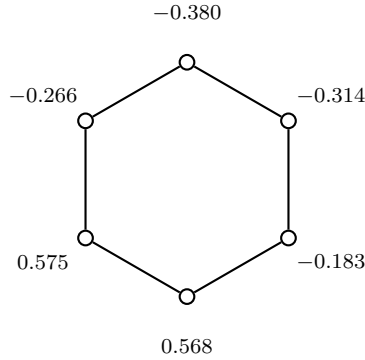


Figure 9: A first positive eigenfunction on the cycle of Example 4.3.

Example 4.4. Let G be the path $v_0 - v_1 - v_2 - v_3 - v_4$ with $B = V$ and consecutive edge weights $1, 2, \frac{1}{2}, 3$. Then $S = L$, and a first positive eigenfunction is, to three decimals,

$$(0.517, 0.346, 0.203, -0.502, -0.565),$$

as in Figure 10. The values are strictly decreasing. The only local extrema are the endpoints, and there is a single sign change, between v_2 and v_3 .

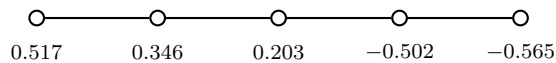


Figure 10: A first positive eigenfunction on the path of Example 4.4.

References

- [1] L. Wang, Generic properties of Steklov eigenfunctions, *Trans. Amer. Math. Soc.* **375** (2022), 8241–8255.
- [2] K. Uhlenbeck, Generic properties of eigenfunctions, *Amer. J. Math.* **98** (1976), 1059–1078.
- [3] B. Hua, Y. Huang and Z. Wang, First eigenvalue estimates of Dirichlet-to-Neumann operators on graphs, *Calc. Var. Partial Differential Equations* **56** (2017), 178.
- [4] L. Chen, B. Hua and Y. Shi, Generic properties of discrete Steklov eigenfunctions, arXiv:2607.20267, 2026.
- [5] J. Sun, Multiple eigenvalue sensitivity analysis, *Linear Algebra Appl.* **137/138** (1990), 183–211.
- [6] H. Federer, *Geometric measure theory*, Springer, 1969.
- [7] M. Fiedler, A property of eigenvectors of nonnegative symmetric matrices and its application to graph theory, *Czechoslovak Math. J.* **25** (1975), 619–633.
- [8] O. Knill, A graph theoretical Poincaré–Hopf theorem, arXiv:1201.1162, 2012.
- [9] T. Banchoff, Critical points and curvature for embedded polyhedra, *J. Differential Geom.* **1** (1967), 245–256.
- [10] R. Forman, Morse theory for cell complexes, *Adv. Math.* **134** (1998), 90–145.
- [11] G. Berkolaiko, A lower bound for nodal count on discrete and metric graphs, *Comm. Math. Phys.* **278** (2008), 803–819.
- [12] Y. Colin de Verdière, Magnetic interpretation of the nodal defect on graphs, *Anal. PDE* **6** (2013), 1235–1242.
- [13] F. R. Gantmacher and M. G. Krein, *Oscillation Matrices and Kernels and Small Vibrations of Mechanical Systems*, revised English ed., AMS Chelsea, 2002.